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(54) **SELECTED AAV COMPOSITIONS HAVING
PREFERRED BRAIN ENRICHMENT**

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ABSTRACT

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Described herein are compositions and kits comprising recombinant adeno-associated viruses (rAAVs) with increased viral transduction enrichment in the CNS. The rAAV compositions described herein encapsidate a transgene, such as a therapeutic nucleic acid. Gene therapy using the rAAVs is described. Also described are methods of treating CNS-related diseases and conditions.

Specification includes a Sequence Listing.

AA sequence	SEQ ID	Dna Sequence	SEQ ID	Enrich Avg #1	Enrich Avg #2
AQANTVRNTAQ	2	GCTCAGGCTAATACTGTTGTAATACTGCTCAG	67	-0.434	0.531
AQDNHVRTVAQ	3	GCTCAGGATAATCATGTTGCTACTGTTGCTCAG	68	-0.763	0.492
AQENSTKSMQAQ	4	GCTCAGGAGAACTTCTACTAAGTCTTTGGCTCAG	69	-2.409	0.232
AQENTVRNMAQ	5	GCTCAGGAGAACTGTTGTAATATGGCTCAG	70	-0.555	0.489
AQGNAIRNGAQ	6	GCTCAGGGTAATGCTATTGTAATGGTGCTCAG	71	-0.211	-0.095
AQGNNAVKPFAQ	7	GCTCAGGGTAATGCTGTTAAGCCTTTGCTCAG	72	-1.316	0.095
AQGNNGTRPTAQ	8	GCTCAGGGTAATGGTACTCGTTTGTGCTCAG	73	-1.262	-2.109
AQHNNVRPVAQ	9	GCTCAGCATAATAATGTTGCTCCTGTTGCTCAG	74	-0.721	0.577
AQLLRDKSSAQ	10	GCTCAGTTGTTGCGTGATAAGTCTTCTGCTCAG	75	-1.735	-2.33
AQMNNNAVLAQ	11	GCTCAGATGAATAATACTGCTGTTTTGGCTCAG	76	-2.18	-1.749
AQNNTTKLAAQ	12	GCTCAGAATAATACTACTAAGTTGGCTGCTCAG	77	-1.585	-2.091
AQNNTVKNLAAQ	13	GCTCAGAATAATACTGTTAAGAATTTGGCTCAG	78	-0.394	0.957
AQSNVIRNSAQ	14	GCTCAGTCTAATGTTATTGTAATTTCTGCTCAG	79	0.109	1.087
AQTLKSPATAQ	15	GCTCAGACTTTGAAGTCTCCTGCTACTGCTCAG	80	-2049	-1.551
AQTN5VKLIAQ	16	GCTCAGACTAATTCTGTTAAGGCTACTGCTCAG	81	-2.318	0.338
AQTSVVPVGAQ	17	GCTCAGACTTCTGTTGTTCTGTTGGTGCTCAG	82	-1.797	-2.151
AQYNSTKNMAQ	18	GCTCAGTATAATTCTACTAAGAATATGGCTCAG	83	-2.23	0.692
AQANSVKNIAQ	19	GCTCAGGCTAATTCTGTTAAGAATATTGCTCAG	84	-1.605	1.117
AQLNTTKPIDG	20	GCTCAGTTGAATACTACTAAGCCTATTGCTCAG	85	-1.914	0.864

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AQGNAIRNGAQ	6	GCTCAGGTAATGCTATTGTAATGGTCTCAG	71	-0.211	-0.095
AQGNNAVXPPAQ	7	GCTCAGGTAATGCTGTTAAGCCTTTTGCTCAG	72	-1.316	0.095
AQNGNGTRPTAQ	8	GCTCAGGGTAATGGTACTGCTTTGTTGCTCAG	73	-1.862	-2.109
AQHNNVRPVAQ	9	GCTCAGCATAATAATGTTGTTGCTGTTGCTCAG	74	-0.721	0.577
AQLLRDKSSAQ	10	GCTCAGTTGTTGGTGATAAGTCTTCTGCTCAG	75	-1.735	-2.33
AQMNNNTAVLAQ	11	GCTCAGATGAATAATACTGCTGTTTGGCTCAG	76	-2.18	-1.749
AQNNTTKLAAQ	12	GCTCAGAAATAATACTACTAAGTTGGCTGCTCAG	77	-1.585	-2.091
AQNNTVXNLAQ	13	GCTCAGAAATAATACTGTTAAGAAATTTGGCTCAG	78	-0.394	0.957
AQSNVIRNSAQ	14	GCTCAGTCTAATGTTATTGTAATTCGCTCAG	79	0.109	1.087
AQTLKSPATAQ	15	GCTCAGACTTTGAAGTCTCCTGCTACTGCTCAG	80	-2049	-1.551
AQTNSVKLIAQ	16	GCTCAGACTAATCTGTTAAGGCTACTGCTCAG	81	-2.318	0.338
AQTSVVVPVGAQ	17	GCTCAGACTTCTGTTGTTCCCTGTTGGTCTCAG	82	-1.797	-2.151
AQYNSTKNMMAQ	18	GCTCAGTATAATTCTACTAAGAATAATGGCTCAG	83	-2.23	0.692
AQANSVKNNIAQ	19	GCTCAGGCTAATTCGTGTTAAGAAATAATGCTCAG	84	-1.605	1.117
AQLNTTKPIDG	20	GCTCAGTTGAATACTACTAAGCCTATTGCTCAG	85	-1.914	0.864

FIG. 1

SELECTED AAV COMPOSITIONS HAVING PREFERRED BRAIN ENRICHMENT

BACKGROUND

[0001] Recombinant adeno-associated viruses (rAAVs) are widely used as vectors for gene delivery in therapeutic applications because of their ability to transduce both dividing and non-dividing cells, their long-term persistence as episomal DNA in infected cells, and their low immunogenicity. These characteristics make them appealing for applications in therapeutic applications, such as gene therapy. However, there is a need to significantly improve the performance of existing AAV serotypes to selectively and efficiently express in distinct cell-types, upon systemic delivery to a subject. This need is especially acute when the AAV must be expressed in the central nervous system (CNS).

SUMMARY OF THE INVENTION

[0002] Disclosed herein are rAAVs with engineered transduction enrichment into the capsid structure through iterative rounds of selection in non-human primates (NHPs), yielding variants having increased transduction enrichment when measured in the CNS.

[0003] The present invention provides rAAVs with widespread transduction to the CNS.

[0004] The present invention provides, in an aspect, a peptide insertion sequence comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1.

[0005] Another aspect of the invention is a modified capsid protein wherein the AAV capsid protein, with a peptide insertion comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1, is characterized by enriched CNS transduction in a subject.

[0006] The present disclosure moreover includes pharmaceutical compositions comprising rAAVs with a peptide insertion comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1, and a pharmaceutically acceptable excipient.

[0007] Aspects disclosed herein provide methods of treating a disease or condition in a subject comprising administering a therapeutically effective amount of a pharmaceutical formulation comprising the AAV capsid protein or the AAV capsid of the present disclosure. In some embodiments, the disease or the condition is a disease or a condition of the CNS, and brain of the subject. Relatedly, the invention includes use of the rAAVs in the manufacture of a medication for treating or preventing the disease or medical condition.

[0008] Other aspects of the invention will be apparent from the detailed description and claims that follow.

BRIEF DESCRIPTION OF THE FIGURES

[0009] The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which:

[0010] FIG. 1 shows AAV capsid protein insertion and substitution amino acid sequences and DNA sequences encoding the amino acid sequences which were found in the non-human primate CNS after two rounds of evaluation of engineered AAV libraries.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0011] In one aspect the disclosure provides rAAVs with high expression levels in the CNS.

[0012] In one aspect, the disclosure provides rAAVs with a peptide insertion and or substitution comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1.

[0013] In some embodiments, the amino acid sequence as described in Table 1, is selected from

AQANTVRNTAQ,	(SEQ ID NO: 2)
AQDNHVRTVAQ,	(SEQ ID NO: 3)
AQENSTKSMAQ,	(SEQ ID NO: 4)
AQENTVRNMAQ,	(SEQ ID NO: 5)
AQGNAIRNGAQ,	(SEQ ID NO: 6)
AQGNVAKPFAQ,	(SEQ ID NO: 7)
AQNGGTRPTAQ,	(SEQ ID NO: 8)
AQHNNVRPVAQ,	(SEQ ID NO: 9)
AQLLRDKSSAQ,	(SEQ ID NO: 10)
AQMNTAVLAQ,	(SEQ ID NO: 11)
AQNNTTKLAAQ,	(SEQ ID NO: 12)
AQNNTVKNLAAQ,	(SEQ ID NO: 13)
AQSNVIRNSAQ,	(SEQ ID NO: 14)
AQTLKSPATAQ,	(SEQ ID NO: 15)
AQTSNVKLIAQ,	(SEQ ID NO: 16)
AQTSVVPVGAQ,	(SEQ ID NO: 15)
AQYNSTKNMAQ,	(SEQ ID NO: 18)
AQANSVKNIAQ, and	(SEQ ID NO: 19)
AQLNNTTKPIDG.	(SEQ ID NO: 20)

[0014] In some embodiments, the insertion sequence as described in Table 1, is selected from

ANTVRNT,	(SEQ ID NO: 21)
DNHVRTV,	(SEQ ID NO: 22)
ENSTKSM,	(SEQ ID NO: 23)
ENTVRNM,	(SEQ ID NO: 24)
GNAIRNG,	(SEQ ID NO: 25)
GNAVKPF,	(SEQ ID NO: 26)
GNGTRPT,	(SEQ ID NO: 27)
HNNVRPV,	(SEQ ID NO: 28)
LLRDKSS,	(SEQ ID NO: 29)
MNNTAVL,	(SEQ ID NO: 30)
NNTTKLA,	(SEQ ID NO: 31)
NNTVKNL,	(SEQ ID NO: 32)
SNVIRNS,	(SEQ ID NO: 33)
TLKSPA,	(SEQ ID NO: 34)
TNSVKLI,	(SEQ ID NO: 35)
TSVVPVG,	(SEQ ID NO: 36)
YNSTKNM, and	(SEQ ID NO: 37)
ANSVKNI.	(SEQ ID NO: 38)

[0015] In some embodiments, the insertion sequence is represented by the peptide sequences listed in Table 1.

TABLE 1			
Sequence	SEQ ID NO	Sequence	SEQ ID NO
ANTVRNT	21	AQANTVRNTAQ	2
DNHVRTV	22	AQDNHVRTVAQ	3
ENSTKSM	23	AQENSTKSMAQ	4
ENTVRNM	24	AQENTVRNMAQ	5
GNAIRNG	25	AQGNAIRNGAQ	6
GNAVKPF	26	AQGNAVKPFAQ	7

TABLE 1-continued

Sequence	SEQ ID NO	Sequence	SEQ ID NO
GNGTRPT	27	AQNGNGTRPTAQ	8
HNNVRPV	28	AQHNNVRPVAQ	9
LLRDKSS	29	AQLLRDKSSAQ	10
MNNTAVL	30	AQMNTAVLAQ	11
NNTTKLA	31	AQNNTTKLAAQ	12
NNTVKNL	32	AQNNTVKNLAQ	13
SNVIRNS	33	AQSNVIRNSAQ	14
TLKSPAT	34	AQTLKSPATAQ	15
TNSVKLI	35	AQTNSVKLIAQ	16
TSVVPVG	36	AQTSVVPVGAQ	17
YNSTKNM	37	AQYNSTKNMAQ	18
ANSVKNI	38	AQANSVKNIAQ	19
LNTTKPI	39	AQLNNTTKPIDG	20

[0016] In some aspects, the insertion amino acid sequence is at least 71.4% identical to the amino acid sequence provided in Table 1 and FIG. 1. In some aspects, the insertion amino acid sequence is at least 86.7% identical to the amino acid sequence provided in Table 1 and FIG. 1.

[0017] Also disclosed herein are methods and kits for producing therapeutic recombinant AAV (rAAV) particles, as well as methods and pharmaceutical compositions or formulations comprising the rAAV particles, for the treatment of a disease or condition affecting the CNS.

[0018] Disclosed herein are AAV capsids engineered with increased viral transduction in the CNS. The AAV capsids can encapsidate a viral vector with a heterologous nucleic acid encoding, for example, a therapeutic gene expression product. Transduction of the heterologous nucleic acid in the CNS can be achieved upon systemic delivery to a subject of the AAV capsid of the present disclosure encapsidating a heterologous nucleic acid. The AAV capsids disclosed herein are advantageous for many applications of gene therapy to treat human disease, including, but not limited to, disorders of the central nervous system.

[0019] The recombinant AAV vectors comprising a nucleic acid sequence encoding the AAV capsid proteins of the present disclosure as also provided herein. For example, the viral vectors of the present disclosure comprise a nucleic acid sequence comprising the AAV viral Cap (Capsid) encoding VP1, VP2, and VP3, at least one of which is modified to produce the AAV capsid proteins of the present disclosure. The recombinant AAV vector provided can be derived from an AAV serotype (e.g., AAV9) or a variant AAV serotype including an insertion of the present invention.

AAV Capsids

[0020] Provided herein are modified adeno-associated (AAV) virus capsid compositions useful for integrating a transgene into a target cell or environment (in a subject when they are administered systemically to the subject).

[0021] An rAAV comprises an AAV capsid that can be engineered to encapsidate a heterologous nucleic acid (e.g., therapeutic nucleic acid, gene editing machinery). The AAV capsid is made up of three AAV capsid protein monomers, VP1, VP2, and VP3. Sixty copies of these three VP proteins interact in a 1:1:10 ratio to form the viral capsid. VP1 covers the whole of VP2 protein in addition to a ~137 amino acid N-terminal region (VPlu), VP2 covers the whole of VP3 in addition to ~65 amino acid N-terminal region (VP1/2 common region). The three capsid proteins share a conserved amino acid sequence of VP3, which in some cases is the region beginning at amino acid position 138 (e.g., AA139-736).

[0022] While not wishing to be bound by theory, it is understood that a parent AAV capsid sequence comprises a VP1 region. In certain embodiments, a parent AAV capsid sequence comprises a VP1, VP2 and/or VP3 region, or any combination thereof. A parent VP1 sequence may be considered synonymous with a parent AAV capsid sequence.

[0023] The AAV VP3 structure contains highly conserved regions that are common to all serotypes, a core eight-stranded β -barrel motif (β B- β I) and a small α -helix (α A). The loop regions inserted between the β -strands consist of the distinctive HI loop between β -strands H and I, the DE loop between β -strands D and E, and nine variable regions (VRs), which form the top of the loops. These VRs, such as the AA588 loop, are found on the capsid surface and can be associated with specific functional roles in the AAV life cycle including receptor binding, transduction and antigenic specificity.

[0024] In some aspects, the rAAV variant of the present invention comprises an AAV capsid protein having a peptide insertion at the residues corresponding to amino acids 588-589 of the AAV9 native sequence of SEQ ID NO: 1.

[0025] The AAV capsids comprise AAV capsid proteins (e.g., VP1, VP2, and VP3), each with an insertion, such as in the 588 loop of a parental AAV capsid protein structure (AAV9 VP1 numbering). The 588 loop contains the site of heparan sulfate binding of AAV2 and is amenable to peptide display. The only known receptors for AAV9 is N-linked terminal galactose and AAV receptor (AAVR), but many indications point toward there being others. Modifications to AAV9 588 loop are shown herein to confer an increased transgene transduction in target in vivo environments.

[0026] The present invention provides, in an aspect, a peptide insertion at the AAV 588 loop comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1.

[0027] Disclosed herein are AAV capsids comprising AAV capsid proteins with an insertion at the 588 loop that confer a higher enrichment for transduction in CNS cell types (e.g., brain endothelial cells, neurons, astrocytes). In particular, the AAV capsid proteins disclosed herein enable rAAV-mediated transduction of a heterologous nucleic acid (e.g., transgene) in the CNS of a subject. The AAV capsids of the present disclosure may be formulated as a pharmaceutical composition. In addition, the AAV capsids can be isolated and purified to be used for a variety of applications.

[0028] In some embodiments, the rAAV capsid of the present disclosure are generated using the methods disclosed herein. In some instances, the rAAV capsid is chimeric. In some instances, the rAAV, or variant AAV protein comprises

therein, confer an increase in a localization of the rAAV within the target tissue, as compared to the parental AAV capsid or capsid protein.

AAV Capsid Proteins

[0029] Disclosed herein are recombinant AAV (rAAV) capsids which comprise AAV capsid proteins that are engineered with a modified capsid protein (e.g., VP1, VP2, VP3). In some embodiments, the rAAV capsid proteins of the present disclosure are generated using the methods disclosed herein. In some embodiments, the AAV capsid proteins are used in the methods of delivering a therapeutic nucleic acid (e.g., a transgene) to a subject. In some instances, the rAAV capsid proteins have desired AAV expression rendering them particularly suitable for certain therapeutic applications, e.g., the treatment of a disease or disorder in a subject such as those disclosed herein.

[0030] The rAAV capsid proteins are engineered for optimized expression in the CNS, for example the brain, of a subject upon systemic administration of the rAAV to the subject. The rAAV capsid proteins are engineered to include the insertions provided in Table 1 and FIG. 1. The rAAV capsid proteins including the insertions provided in Table 1 and FIG. 1 are engineered to achieve efficient transduction of an encapsidated transgene. In particular, the rAAV capsid proteins have an increased expression enrichment in the brain of a subject.

[0031] The engineered AAV capsid proteins described herein have, in some cases, an insertion of an amino acid that is heterologous to the parental AAV capsid protein at amino acid positions in the 588 loop. In some embodiments, the amino acid is not endogenous to the parental AAV capsid protein at the amino acid position of the insertion. The amino acid may be a naturally occurring amino acid in the same or equivalent amino acid position as the insertion of the substitution in a different AAV capsid protein.

[0032] Generally, the insertion comprises a five-, six-, or seven-amino acid sequence (5-mer, 6-mer, or 7-mer, respectively) that is inserted or substituted at the 588 loop in a parental AAV capsid protein. Aspects provided herein provide amino acid insertions comprising seven amino acid polymer (7-mer) inserted at AA588-589, and may additionally include a substitution of one or two amino acids at amino acid positions flanking the 7-mer sequence (e.g., AA587-588 and/or AA589-590) to produce an eleven amino acid polymer (11-mer) at the 588 loop of a parental AAV capsid protein. The 7-mers described herein were advantageously generated using polymerase chain reaction (PCR) with degenerate primers, where each of the seven amino acids is encoded by a deoxyribose nucleic acid (DNA) sequence N-N-K. "N" is any of the four DNA nucleotides and K is guanine (G) or thymine (T). This method of generating random 7-mer amino acid sequences enables 1.28 billion possible combinations at the protein level.

[0033] The rAAV capsid proteins of the present disclosure comprise an insertion of an amino acid in an amino acid sequence of an AAV capsid protein. The AAV capsid, from which an engineered AAV capsid protein of the present disclosure is produced, is referred to as a "parental" AAV capsid. The complete genome of AAV-1 is provided in GenBank Accession No. NC_002077; the complete genome of AAV-2 is provided in GenBank Accession No. NC_001401 and Srivastava et al., J. Virol., 45:555-564 (1983); the complete genome of AAV-3 is provided in

GenBank Accession No. NC_1829; the complete genome of AAV-4 is provided in GenBank Accession No. NC_001829; the AAV-5 genome is provided in GenBank Accession No. AF085716; the complete genome of AAV-6 is provided in GenBank Accession No. NC_001862; at least portions of AAV-7 and AAV-8 genomes are provided in GenBank Accession Nos. AX753246 and AX753249, respectively; the AAV-9 genome is provided in Gao et al., J. Virol., 78:6381-6388 (2004); the AAV-10 genome is provided in Mol. Ther., 13 (1): 67-76 (2006); the AAV-11 genome is provided in Virology, 330 (2): 375-383 (2004); portions of the AAV-12 genome are provided in Genbank Accession No. DQ813647; portions of the AAV-13 genome are provided in Genbank Accession No. EU285562.

[0034] In some cases, the parental AAV is derived from an AAV with a serotype selected from AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11 and AAV12. The AAV capsid protein that is “derived” from another may be a variant AAV capsid protein. A variant may include, for example, a heterologous amino acid in an amino acid sequence of the AAV capsid protein. The heterologous amino acid may be non-naturally occurring in the AAV capsid protein. The heterologous amino acid may be naturally occurring in a different AAV capsid protein. In some instances, the parental AAV capsid is described in US Pat Publication 2020/0165576 and U.S. Pat. App. Ser. No. 62/832,826 and PCT/US20/20778; the content of each of which is incorporated herein.

[0035] In some instances, the parental AAV is AAV9. In some instances, the amino acid sequence of the AAV9 capsid protein comprises SEQ ID NO: 1. The amino acid sequence of AAV9 VP1 capsid protein (>tr|Q6JC40|Q6JC40_9VIRU Capsid protein VP1 OS=Adeno-associated virus 9 OX=235455 GN=cap PE=1 SV=1) is provided in SEQ ID NO: 1 (MAADGYLPDWLEDNLSEGIREWWALKP-GAPQPKANQHQDNARGVLPLPGYKYLPGP NGLDKGEPVNAADAAALEHDKAYDQQLK-AGDNPYLKYNHADAEEFQERLKEDTSFGG NLGRAVFQAKKRLLEPLGLVEEAAK-TAPGKKRPVEQSPQEPDSSAGIGKSGAQPAKKRL NFGQTGDTEVPDPQPIGEPPAAPSGVGSLSMASGG-GAPADNNEGADGVGSSSGNWH CDSQWLGDRL VITSTRTWALPTYNNHLYKQISNSTSGGSSNDNAY-FGYSTPWGYFDNF RFHCHFSRDLWQRLINNNWG-FRPKRLNFKLFNIQVKEVTDNNGVKTIANNLT-STVQVFT DSDYQLPYVLGSAHEGCLPPFPAD-VFMIPQYGYLTLNDGSQAVGRSSFYCYEFP SQMLRTGNNFQFSYEFENVPFHSSYAHQSGLDRLMN-PLIDQYLYLSKTINGSGQNQQLKFSV AGPSN-MAVQGRNYIPGPSYRQQRVSTTVTQNNNSEFAWP-GASSWALNGRNSLMNPGP AMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDAD-KVMITNEEEIKTTNPVATESYGGQVA TNHQAQAQAQTGWVQNQGIL-PGMVWQDRDVYLQGPWAKIPH-TDGNFHPSPMLGGF GMKHPPPQILIKNTPV-PADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKEN SKRWNP E IQYTSNYYKSNNVEFAVNTEGVY-SEPRPIGTRYLTRNL). In some instances, the parental AAV capsid protein sequence is 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% homologous to SEQ ID NO: 1.

[0036] AAV capsid proteins from native AAV serotypes, such as AAV9, with tropisms including the liver activate the innate immune response, which in some cases causes a severe inflammatory response in a subject, which can lead to multi-organ failure. By improving transduction enrichment of a native AAV serotype for a target in vivo tissue (e.g., brain), the rAAV particles of the present disclosure reduce the immunogenic properties of AAV-mediated transgene delivery and prevent activation of the innate immune response.

[0037] In some instances, the parental AAV capsid protein comprises the entire VP1 region provided in SEQ ID NO: 1 (e.g., amino acids 1-736). In some instances, the parental AAV capsid protein comprises amino acids 217-736 in SEQ ID NO: 1, which is the common region found in VP1, VP2 and VP3 AAV9 capsid proteins. In some instances, the AAV capsid protein comprises amino acids 64-736 in SEQ ID NO: 1, which is the common region found in VP1 and VP2. The parental AAV capsid protein sequence may comprise amino acids selected from 1-736, 10-736, 20-736, 30-736, 40-736, 50-736, 60-736, 70-736, 80-736, 90-736, 100-736, 110-736, 120-736, 130-736, 140-736, 150-736, 160-736, 170-736, 180-736, 190-736, 200-736, 210-736, 220-736, 230-736, 240-736, 250-736, 260-736, 270-736, 280-736, 290-736, 300-736, 310-736, 320-736, 330-736, 340-736, 350-736, 360-736, 370-736, 380-736, 390-736, 400-736, 410-736, 420-736, 430-736, 440-736, and 450-736, from SEQ ID NO: 1. In some aspects, the rAAV variant comprises an AAV capsid protein comprising an amino acid sequence that is at least 98% identical to amino acid 217 to amino acid 736 of SEQ ID NO: 1. In some instances, the amino acid insertion is at a three (3)-fold axis of symmetry of a corresponding parental AAV capsid protein.

[0038] Disclosed herein are insertions of an amino acid sequence in an AAV capsid protein. Where the sequence numbering designation “588-589” is noted for AAV9, for example AAV VP1, the invention also includes insertions in similar locations in the other AAV serotypes. As used herein, “AA588-589” indicates that the insertion of the amino acid (or amino acid sequence) is immediately after an amino acid (AA) at position 588 and immediately before an AA at position 589 within an amino acid sequence of a parental AAV VP capsid protein (VP1 numbering). Amino acids 587-591 include a motif comprising “AQAQA” as set forth in SEQ ID NO: 1. Exemplary AAV capsid protein sequences are provided in Table 2. For example, ANTVRNT (SEQ ID NO: 21) is inserted at AA588-589 in an AAV9 capsid amino acid sequence, and provides variant A (SEQ ID NO: 45). It is envisioned that the insertions disclosed herein (Table 1 and FIG. 1) may be inserted at AA588-589 in an amino acid sequence of a parental AAV9 capsid protein, a variant thereof, or equivalent amino acid position of a parental AAV of a different serotype (e.g., AAV1, AAV2, AAV3, and the like). In any AAV capsid protein sequence disclosed herein, the amino acid at position 449 may be R or K.

TABLE 2

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
40	AAV-PHP-eB	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGDESVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSOMLRTGNFQFSYEFENVPFH SSYAHSQLDLRLMNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS DGTLAVPFKAQAQ TGWVQNGGILPGMVWQDR RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPQILIKNTTPVPADPPTAFNKDKLNSFITQYSTG QVSVIEIWEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTGVSSEPRPIGTRYLTRNL
41	AAV .CAP-A4	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGDESVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSOMLRTGNFQFSYEFENVPFH SSYAHSQLDLRLMNPLIDQYLYLSRTI KDNT P GRQ TLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQAQAQ TGWVQNGGILPGMVWQDRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPPQILIKN TPVPADPPTAFNKDKLNSFITQYSTGQVSVIEI EWELQKENS KRWNPEIQYTSNYYKSNVEFAVNT GVYSEPRPIGTRYLTRNL
42	AAV .CAP-B2	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGDESVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSOMLRTGNFQFSYEFENVPFH SSYAHSQLDLRLMNPLIDQYLYLSRTI QGGKQ SVQ TLKFSVAGPSNMAVQGRNYIPGPSYRQQR STTVTQNNSEFAWPGASSWALNGRNSLMNP PAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV ADKVMITNEEEIKTTNPVATESYGQVATNHQS GTLAVPFKAQAQ TGWVQNGGILPGMVWQDRD VYLQGPWAKIPHTDGNFHPSPLMGGFGMKHPP PQILIKNTTPVPADPPTAFNKDKLNSFITQYSTGQ VSVIEIWEWELQKENS KRWNPEIQYTSNYYKSNVE FAVNTGVSSEPRPIGTRYLTRNL

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
43	AAV .CAP-B10	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGDESVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSOMLRTGNFQFSYEFENVPFH SSYAHSQLDLRLMNPLIDQYLYLSRTI DGAAT KNQ TLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS DGTLAVPFKAQAQ TGWVQNGGILPGMVWQDR DVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHP PQILIKNTTPVPADPPTAFNKDKLNSFITQYSTG QVSVIEIWEWELQKENS KRWNPEIQYTSNYYKSNV EFAVNTGVSSEPRPIGTRYLTRNL
44	AAV .CAP-B22	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGDESVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSOMLRTGNFQFSYEFENVPFH SSYAHSQLDLRLMNPLIDQYLYLSRTI DGQSS KSQ TLKFSVAGPSNMAVQGRNYIPGPSYRQQR STTVTQNNSEFAWPGASSWALNGRNSLMNP PAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV ADKVMITNEEEIKTTNPVATESYGQVATNHQS GTLAVPFKAQAQ TGWVQNGGILPGMVWQDRD VYLQGPWAKIPHTDGNFHPSPLMGGFGMKHPP PQILIKNTTPVPADPPTAFNKDKLNSFITQYSTGQ VSVIEIWEWELQKENS KRWNPEIQYTSNYYKSNV FAVNTGVSSEPRPIGTRYLTRNL
45	Variant A	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGDESVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSOMLRTGNFQFSYEFENVPFH SSYAHSQLDLRLMNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQANTVRNTAQAQ TGWVQNGGILPGMVWQDR RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPQILIKNTTPVPADPPTAFNKDKLNSFITQYSTG

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
46	Variant B	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPNGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFAQKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGESVDPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFDNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPS QMLRTGNMFQFSYEFENVFPFH SSYAHQSLSLRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQDNHVRTVAQAQTGWVQNGQILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPPPQILIKNTTPVPADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
47	Variant C	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPNGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFAQKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGESVDPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFDNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPS QMLRTGNMFQFSYEFENVFPFH SSYAHQSLSLRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQENSTKSMAQAQTGWVQNGQILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPPPQILIKNTTPVPADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
48	Variant D	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPNGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFAQKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGESVDPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFDNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPS QMLRTGNMFQFSYEFENVFPFH SSYAHQSLSLRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQENTVRNMAQAQTGWVQNGQILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		HPPPPQILIKNTTPVPADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
49	Variant E	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPNGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFAQKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGESVDPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFDNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPS QMLRTGNMFQFSYEFENVFPFH SSYAHQSLSLRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQLNTTKPIDGAQTGWVQNGQILPGMVWQDR DVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTTPVPADPPTAFNKDKLNSFITQYSTG QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNV EFAVNTEGVYSEPRPIGTRYLTRNL
50	Variant F	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPNGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFAQKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGESVDPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFDNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPS QMLRTGNMFQFSYEFENVFPFH SSYAHQSLSLRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQGNAINRGAQAQTGWVQNGQILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPPPQILIKNTTPVPADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
51	Variant G	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPNGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEPQERLKEDTSFGGNLGRAVFAQKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGESVDPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFDNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPS QMLRTGNMFQFSYEFENVFPFH SSYAHQSLSLRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQSNVIRNSAQAQTGWVQNGQILPGMVWQDR

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		DVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTVPADPPTAFNKDKLNSFITQYSTG QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNV EFAVNTGEGVYSEPRPIGTRYLTRNL
52	Variant H	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAATAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTVSDPDPQPIGEPPA APSGVGSLTMSAGGAPVADNNEGADGVGSSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQANSVKINIAQA QGTGWVQNGGILPGMVWQDR DVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTVPADPPTAFNKDKLNSFITQYSTG QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNV EFAVNTGEGVYSEPRPIGTRYLTRNL
53	Variant I	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAATAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTVSDPDPQPIGEPPA APSGVGSLTMSAGGAPVADNNEGADGVGSSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQYNSTKNMAQA QGTGWVQNGGILPGMVWQD RDVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMK HPPQILIKNTVPADPPTAFNKDKLNSFITQYST TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTGEGVYSEPRPIGTRYLTRNL
54	Variant J	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAATAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTVSDPDPQPIGEPPA APSGVGSLTMSAGGAPVADNNEGADGVGSSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		AQTLKSPATAQA QGTGWVQNGGILPGMVWQDR DVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTVPADPPTAFNKDKLNSFITQYSTG QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNV EFAVNTGEGVYSEPRPIGTRYLTRNL
55	Variant K	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAATAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTVSDPDPQPIGEPPA APSGVGSLTMSAGGAPVADNNEGADGVGSSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQGNVAKPFAQA QGTGWVQNGGILPGMVWQD RDVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMK HPPQILIKNTVPADPPTAFNKDKLNSFITQYST TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTGEGVYSEPRPIGTRYLTRNL
58	Variant L	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAATAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTVSDPDPQPIGEPPA APSGVGSLTMSAGGAPVADNNEGADGVGSSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQGNTRPPTAQA QGTGWVQNGGILPGMVWQD RDVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMK HPPQILIKNTVPADPPTAFNKDKLNSFITQYST TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTGEGVYSEPRPIGTRYLTRNL
59	Variant M	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLPGPGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFPQAKKRLL EPLGLVEEAATAKTPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTVSDPDPQPIGEPPA APSGVGSLTMSAGGAPVADNNEGADGVGSSS GNWHCDSQLGDRVITTTSTRTWALPTYNNHLY KQISNSTSGGSSNDNAYFGYSTPWGYDFNRFH CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK EVTDDNNGVKTIANNLTSTVQVFTDSYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDNGSQAV GRSSFYCLEYFPPSQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRLMNPIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMVQGRNYIPGPSYRQQR VSTTVTQNNSEFAWPAGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQHNNVRPVA QAQTGWVQNQGILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPPPQILIKNTVPVADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
60	Variant N	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLGPNGGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFOAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTESVDPDPQIGEP APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRIITSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFNRFH CHFSRWDQRLINNNWGFPRKRLNFKLFNIQVK EVTDNNGVKTIANNLSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDGSAV GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPPH SSYAHQSQSLDRMLNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQLLRDKSSA QAQTGWVQNQGILPGMVWQDR DVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTVPVADPPTAFNKDKLNSFITQYSTG QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNV EFAVNTEGVYSEPRPIGTRYLTRNL
61	Variant O	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLGPNGGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFOAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTESVDPDPQIGEP APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRIITSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFNRFH CHFSRWDQRLINNNWGFPRKRLNFKLFNIQVK EVTDNNGVKTIANNLSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDGSAV GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPPH SSYAHQSQSLDRMLNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQMNTAVLA QAQTGWVQNQGILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPPPQILIKNTVPVADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
62	Variant P	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLGPNGGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFOAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTESVDPDPQIGEP APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRIITSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFNRFH CHFSRWDQRLINNNWGFPRKRLNFKLFNIQVK EVTDNNGVKTIANNLSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDGSAV GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPPH SSYAHQSQSLDRMLNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNNEFAWPGASSWALNGRNSLMNP

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQNNTTKLAA QAQTGWVQNQGILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK THPPPPQILIKNTVPVADPPTAFNKDKLNSFITQYS GQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN
63	Variant Q	NVEFAVNTEGVYSEPRPIGTRYLTRNL MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLGPNGGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFOAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTESVDPDPQIGEP APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRIITSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFNRFH CHFSRWDQRLINNNWGFPRKRLNFKLFNIQVK EVTDNNGVKTIANNLSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDGSAV GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPPH SSYAHQSQSLDRMLNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQNNTVKNLA QAQTGWVQNQGILPGMVWQD RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK HPPPPQILIKNTVPVADPPTAFNKDKLNSFITQYS TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
64	Variant R	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLGPNGGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFOAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTESVDPDPQIGEP APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRIITSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFNRFH CHFSRWDQRLINNNWGFPRKRLNFKLFNIQVK EVTDNNGVKTIANNLSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDGSAV GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPPH SSYAHQSQSLDRMLNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR VSTTVTQNNNEFAWPGASSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEEIKTTNPVATESYGQVATNHQS AQSNVIRNSA QAQTGWVQNQGILPGMVWQDR DVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTVPVADPPTAFNKDKLNSFITQYSTG QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNV EFAVNTEGVYSEPRPIGTRYLTRNL
65	Variant S	MAADGYLPDWLEDNLSEGIREWALKPGAPQP KANQQHQDNARGLVLPGYKYLGPNGGLDKGE PVNAADAAALEHDKAYDQQLKAGDNPYLKYN HADAEFQERLKEDTSFGGNLGRAVFOAKKRLL EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTDGTESVDPDPQIGEP APSGVGSLTMASGGGAPVADNNEGADGVGSS GNWHCDSQWLGDRIITSTRTWALPTYNNHLY KQISNSTSGSSNDNAYFGYSTPWGYDFNRFH CHFSRWDQRLINNNWGFPRKRLNFKLFNIQVK EVTDNNGVKTIANNLSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIPQYGYLTLDGSAV GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPPH SSYAHQSQSLDRMLNPLIDQYLYLSRTINGSGQ NQOTLKFSVAGPSNMAVQGRNYIPGPSYRQQR

TABLE 2-continued

Exemplary AAV Capsid Protein Sequences		
SEQ ID NO:	Identifier	Sequence
		VSTTVTQNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEIKTTNPVATESYGQVATNHQS AQTNSVKLIA QAQTGWVQNQGILPGMVWQDR DVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMKHP PPQILIKNTPVPADPPTAFNKKDKLNSFITQYSTG QVSVEIEWELQKENSkrwnPEIQYTSNYYKSNV EFAVNTegvyseprPIgTRYLTRLN
66	Variant T	MAADGYLPDWLEDNLSEGIREWwALKPGAPQP KANQQHQDNARGLVLPgyKYLGPNGLDKGE PVNAADAAAEHDKAYDQQLKAGDNpyLKYN HADAEFQERLKEDTSFGGNLGRAVfQAKRLL EPLGLVEEAkTAPGKKRPVEQSPQEPDSSAGIG KSGAQPAKKRLNFGQTGDTEsVPDPQPIGEPPA APSGVGSLTMASGGGAPVADNNEGADGVGSSS GNWHCDsQWLGDVITTTSTRtWALPTYNHLY KQISNSTSGSSNDNAYfGYSTPWgyPDFNRFH CHFSPRDWQRLINNNWGFpKRlNFKLENIQVK EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL GSAHEGCLPPFPADVFMIpQYGYLTlNDGSQAV GRSSFYCLEYFPsQMLRTGNNFQFSYEFENVPFH SSYAHSQSLDRlMNPLIDQYLYLsRTINGSGQ NQQLKFSVAGPSNMAVQGRNYiPGPSYRQQR VSTTVTQNNSEFAWPAGSSWALNGRNSLMNP GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV DADKVMITNEEIKTTNPVATESYGQVATNHQS AQTSVVPVGA QAQTGWVQNQGILPGMVWQD RDVYLQGGPIWAKIPHTDGNFHPSPLMGGFGMK HPPQILIKNTPVPADPPTAFNKKDKLNSFITQYS TGQVSVEIEWELQKENSkrwnPEIQYTSNYYKSN NVEFAVNTegvyseprPIgTRYLTRLN

[0039] The insertions described herein may, in some cases, comprise a 7-mer insertion at AA588-589. It is envisioned that any 7-mer insertion disclosed herein in addition to a substitution with any amino acid at amino acid positions 587-590 [AQAA] may comprise an 11-mer.

[0040] Disclosed herein are AAV capsid proteins with an insertion described above in a parental AAV capsid protein that confers an increased transduction enrichment in the CNS in a subject, even when delivered systemically. One of the many advantages of the AAV capsid proteins described herein is their ability to target tissue and cells within the CNS. The tissue can be the brain. Non-limiting examples of CNS cells include a neuron and a glial cell. Glial cells can be selected from an oligodendrocyte, an ependymal cell, an astrocyte and a microglia.

[0041] In some instances, the AAV capsid protein comprises an insertion of at least or about five, six, or seven amino acids of an amino acid sequence of Table 1 and FIG. 1 at an amino acid position 588-589 in a parental AAV9 capsid protein (SEQ ID NO: 1). In some cases, the AAV capsid protein has increased viral transduction in brain.

[0042] The rAAV capsid proteins of the present disclosure may also have a substitution of an amino acid sequence at amino acid position 452-458 in a parental AAV9 capsid protein, or variant thereof, as described in WO2020068990. In some embodiments, the substitution of the amino acid sequence comprises KDNTpGR (SEQ ID NO: 367) at amino acid position 452-458 in the parental AAV9 capsid protein. In some embodiments, the substitution of the amino

acid sequence comprises DGAATKN (SEQ ID NO: 368) at amino acid position 452-458 in the parental AAV9 capsid protein.

[0043] The rAAV capsid proteins described herein may be isolated and purified. The AAV may be isolated and purified by methods standard in the art such as by column chromatography, iodixanol gradients, or cesium chloride gradients. Methods for purifying AAV from helper virus are known in the art and may include methods disclosed in, for example, Clark et al., Hum. Gene Ther., 10 (6): 1031-1039 (1999); Schenpp and Clark, Methods Mol. Med., 69:427-443 (2002); U.S. Pat. No. 6,566,118 and WO 98/09657.

[0044] In addition, the AAV capsid proteins disclosed herein, either isolated and purified, or not, may be formulated into a pharmaceutical formulation, which in some cases, further comprises a pharmaceutically acceptable carrier.

[0045] The rAAV capsid protein can be conjugated to a nanoparticle, a second molecule, or a viral capsid protein. In some cases, the nanoparticle or viral capsid protein would encapsidate the therapeutic nucleic acid described herein. In some instances, the second molecule is a therapeutic agent, e.g., a small molecule, antibody, antigen-binding fragment, peptide, or protein, such as those described herein.

[0046] “Percent Identity” is the percent of the symbols that actually match. Percent Similarity is the percent of the symbols that are similar. Symbols that are across from gaps are ignored. A similarity is scored when the scoring matrix value for a pair of symbols is greater than or equal to 0.50, the similarity threshold. The scoring matrix used in Version 10 of the Wisconsin Genetics Software Package is BLO-SUM62 (see: Henikoff and Henikoff, (1989) Proc. Natl. Acad. Sci. USA 89:10915).

[0047] Sequence identity/similarity values provided herein can refer to the value obtained using the BLAST+2.5.0 suite of programs using default settings (blast.ncbi.nlm.nih.gov) (Camacho, C., et al. (2009) BLAST+: architecture and applications. BMC Bioinformatics 10:421).

[0048] As those of ordinary skill in the art will understand, BLAST searches assume that proteins can be modeled as random sequences. However, many real proteins comprise regions of nonrandom sequences, which may be homopolymeric tracts, short-period repeats, or regions enriched in one or more amino acids. Such low-complexity regions may be aligned between unrelated proteins even though other regions of the protein are entirely dissimilar. A number of low-complexity filter programs can be employed to reduce such low-complexity alignments. For example, the SEG (Wooten and Federhen, (1993) Comput. Chem. 17:149-63) and XNU (Ci-ayerie and States (1993) Comput. Chem. 17:191-201) low-complexity filters can be employed alone or in combination.

[0049] The terms “substantial identity” and “substantially identical” indicate that a polypeptide or nucleic acid comprises a sequence with between 55-100% sequence identity to a reference sequence, with at least 55% sequence identity, or at least 60%, or at least 65%, or at least 70%, or at least 75%, or at least 80%, or at least 85%, or at least 90%, or at least 95%, or at least 99% sequence identity or any percentage of value within the range of 55-100% sequence identity relative to the reference sequence. The percent sequence identity may occur over a specified comparison window.

Optimal alignment may be ascertained or conducted using the homology alignment algorithm of Needleman and Wunsch, *supra*.

[0050] For example, the insertion sequences may include, but are not limited to, sequences that are not exactly the same as the sequences disclosed herein, but which have, in addition to the substitutions explicitly described for various sequences listed herein, additional substitutions of amino acid residues which substantially do not impair the activity or properties of the sequences described herein, such as those predicted by homology software e.g. BLOSUM62 matrices.

AAV Particles

[0051] The rAAV particles with the insertion sequences described herein have an increased transduction enrichment in the CNS. In some instances, the increased transduction enrichment comprises a 1-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold or 10-fold increase, or more. In some instances, the increased transduction enrichment is at least 1-fold. In some instances, the increased transduction enrichment is at least 2-fold. In some instances, the increased transduction enrichment is at least 4-fold.

[0052] The rAAV particles with the insertion sequences described herein have an increased expression enrichment in the CNS. Detecting whether a rAAV possesses more or less expression includes measuring a level of gene expression product (e.g., RNA or protein) expressed from the heterologous nucleic acid encapsidated by the rAAV in a tissue sample obtained from a subject. Suitable methods for measuring expression of a gene expression product include next-generation sequencing (NGS) and quantitative polymerase chain reaction (qPCR).

[0053] The increased expression in the CNS is represented by the enrichment values provided in FIG. 1.

Heterologous Nucleic Acids

[0054] Disclosed herein are therapeutic nucleic acids useful for the treatment or prevention of a disease or condition, or symptom of the disease or condition. In some embodiments, the therapeutic nucleic acids encode a therapeutic gene expression product. Non-limiting examples of gene expression products include proteins, polypeptides, peptides, enzymes, antibodies, antigen binding fragments, nucleic acid (RNA, DNA, antisense oligonucleotide, siRNA, and the like), and gene editing components, for use in the treatment, prophylaxis, and/or amelioration of the disease or disorder, or symptoms of the disease or disorder. In some instances, the therapeutic nucleic acids are placed in an organism, cell, tissue or organ of a subject by way of a rAAV, such as those disclosed herein.

[0055] Disclosed herein are rAAVs, each comprising a viral vector (e.g., a single stranded DNA molecule (ssDNA)). In some instances, the viral vector comprises two inverted terminal repeat (ITR) sequences that are about 145 bases each, flanking a transgene. In some embodiments, the transgene comprises a therapeutic nucleic acid, and in some cases, a promoter in cis with the therapeutic nucleic acid in an open reading frame (ORF). The promoter is capable of initiating transcription of therapeutic nucleic acid in the nucleus of the target cell. The ITR sequences can be from any AAV serotype. Non-limiting examples of AAV serotypes include AAV1, AAV2, AAV3, AAV4, AAV5, AAV6,

AAV7, AAV8, AAV9, AAV10, AAV11, and AAV12. In some cases, an ITR is from AAV2. In some cases, an ITR is from AAV9.

[0056] Disclosed herein are transgenes that can comprise any number of nucleotides. In some cases, a transgene can comprise less than about 100 nucleotides. In some cases, a transgene can comprise at least about 100 nucleotides. In some cases, a transgene can comprise at least about 200 nucleotides. In some cases, a transgene can comprise at least about 300 nucleotides. In some cases, a transgene can comprise at least about 400 nucleotides. In some cases, a transgene can comprise at least about 500 nucleotides. In some cases, a transgene can comprise at least about 1000 nucleotides. In some cases, a transgene can comprise at least about 5000 nucleotides. In some cases, a transgene can comprise over 5,000 nucleotides. In some cases, a transgene can comprise between about 500 and about 5000 nucleotides. In some cases, a transgene comprises about 5000 nucleotides. In any of the cases disclosed herein, the transgene can comprise DNA, RNA, or a hybrid of DNA and RNA. In some cases, the transgene can be single stranded. In some cases, the transgene can be double stranded.

[0057] Disclosed herein are transgenes useful for modulating the expression or activity of a target gene or gene expression product thereof. In some instances, the transgene is encapsidated by an rAAV capsid protein of an rAAV particle described herein. In some instances, the rAAV particle is delivered to a subject to treat a disease or condition disclosed herein in the subject. In some instances, the delivery is systemic.

[0058] The transgenes disclosed herein are useful for expressing an endogenous gene at a level similar to that of a healthy or normal individual. This is particularly useful in the treatment of a disease or condition related to the under-expression, or lack of expression, of a gene expression product. In some embodiments, the transgenes disclosed herein are useful for overexpressing an endogenous gene, such that an expression level of the endogenous gene is above the expression level of a healthy or normal individual. Additionally, transgenes can be used to express exogenous genes (e.g., active agent such as an antibody, peptide, nucleic acid, or gene editing components). In some embodiments, the therapeutic gene expression product is capable of altering, enhancing, increasing, or inducing the activity of one or more endogenous biological processes in the cell. In some embodiments, the transgenes disclosed herein are useful for reducing expression of an endogenous gene, for example, a dominant negative gene. In some embodiments, the therapeutic gene expression product is capable of altering, inhibiting, reducing, preventing, eliminating, or impairing the activity of one or more endogenous biological processes in the cell. In some aspects, the increase of gene expression refers to an increase by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In one aspect, the protein product of the targeted gene may be increased by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In some aspects, the decrease of gene expression refers to an increase by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In one aspect, the protein product of the targeted gene may be decreased by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%.

[0059] When endogenous sequences (endogenous or part of a transgene) are expressed with a transgene, the endogenous sequences can be full-length sequences (wild-type or mutant) or partial sequences. The endogenous sequences can be functional. Non-limiting examples of the function of these full length or partial sequences include increasing the serum half-life of the polypeptide expressed by a transgene (e.g., therapeutic gene) and/or acting as a carrier.

[0060] A transgene can be inserted into an endogenous gene such that all, some or none of the endogenous gene is expressed. For example, a transgene as described herein can be inserted into an endogenous locus such that some (N-terminal and/or C-terminal to a transgene) or none of the endogenous sequences are expressed, for example as a fusion with a transgene. In other cases, a transgene (e.g., with or without additional coding sequences of the endogenous gene) is integrated into any endogenous locus, for example a safe-harbor locus. For example, a Frataxin (FXN) transgene can be inserted into an endogenous FXN gene. A transgene can be inserted into any gene, e.g., the genes as described herein.

[0061] At least one advantage of the present disclosure is that virtually any therapeutic nucleic acid may be used to express any therapeutic gene expression product. In some instances, the therapeutic gene expression product is a therapeutic protein or a peptide (e.g., antibody, antigen-binding fragment, peptide, or protein). In one embodiment the protein encoded by the therapeutic nucleic acid is between 50-5000 amino acids in length. In some embodiments the protein encoded is between 50-2000 amino acids in length. In some embodiments the protein encoded is between 50-1000 amino acids in length. In some embodiments the protein encoded is between 50-1500 amino acids in length. In some embodiments the protein encoded is between 50-800 amino acids in length. In some embodiments the protein encoded is between 50-600 amino acids in length. In some embodiments the protein encoded is between 50-400 amino acids in length. In some embodiments the protein encoded is between 50-200 amino acids in length. In some embodiments the protein encoded is between 50-100 amino acids in length. In some embodiments the peptide encoded is between 4-50 amino acids in length. In some embodiments, the protein encoded is a tetrapeptide, a pentapeptide, a hexapeptide, a heptapeptide, an octapeptide, a nonapeptide, or a decapeptide. In some embodiments, the protein encoded comprises a peptide of 2-30 amino acids, such as for example 5-30, 10-30, 2-25, 5-25, 10-25, or 10-20 amino acids. In some embodiments, the protein encoded comprises a peptide of at least 11, 12, 13, 14, 15, 17, 20, 25 or 30 amino acids, or a peptide that is no longer than 50 amino acids, e.g., no longer than 35, 30, 25, 20, 17, 15, 14, 13, 12, 11 or 10 amino acids.

[0062] Non-limiting examples of therapeutic protein or peptides include an adrenergic agonist, an anti-apoptosis factor, an apoptosis inhibitor, a cytokine receptor, a cytokine, a cytotoxin, an erythropoietic agent, a glutamic acid decarboxylase, a glycoprotein, a growth factor, a growth factor receptor, a hormone, a hormone receptor, an interferon, an interleukin, an interleukin receptor, a kinase, a kinase inhibitor, a nerve growth factor, a netrin, a neuroactive peptide, a neuroactive peptide receptor, a neurogenic factor, a neurogenic factor receptor, a neuropilin, a neurotrophic factor, a neurotrophin, a neurotrophin receptor, an N-methyl-D-aspartate antagonist, a plexin, a protease, a

protease inhibitor, a protein decarboxylase, a protein kinase, a protein kinase inhibitor, a proteolytic protein, a proteolytic protein inhibitor, a semaphoring, a semaphorin receptor, a serotonin transport protein, a serotonin uptake inhibitor, a serotonin receptor, a serpin, a serpin receptor, and a tumor suppressor. In certain embodiments, the therapeutic protein or peptide is selected from brain-derived neurotrophic factor (BDNF), ciliary neurotrophic factor (CNTF), macrophage colony-stimulating factor (CSF), epidermal growth factor (EGF), fibroblast growth factor (FGF), gonadotropin, interferon-gamma (IFN), insulin-like growth factor 1 (IGF-1), nerve growth factor (NGF), platelet-derived growth factor (PDGF), pigment epithelium-derived factor (PEDF), transforming growth factor (TGF), transforming growth factor-beta (TGF-B), tumor necrosis factor (TNF), vascular endothelial growth factor (VEGF), prolactin, somatotropin, X-linked inhibitor of apoptosis protein 1 (XIAP1), interleukin 1 (IL-1), IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-10, viral IL-10, IL-11, IL-12, IL-13, IL-14, IL-15, IL-16, IL-17, and IL-18.

[0063] A therapeutic gene expression product can comprise gene editing components. Non-limiting examples of gene editing components include those required for CRISPR/Cas, artificial site-specific RNA endonuclease (ASRE), zinc finger endonuclease (ZFN), and transcription factor like effector nuclease (TALEN). In a non-limiting example, a subject having Huntington's disease is identified. The subject is then systemically administered a first amount of a rAAV encapsidating a viral vector encoding ZFN engineered to represses the transcription of the Huntingtin (HTT) gene. The rAAV will include a modified AAV capsid protein that includes an amino acid sequence provided in any one of Table 1 and FIG. 1, so as to allow proper targeting of the ZFN to the nervous system, while reducing expression in off-target organs, such as the liver. If needed, the subject is administered a second or third dose of the rAAV, until a therapeutically effective amount of the ZEN is expressed in the subject's nervous system.

[0064] A therapeutic nucleic acid can comprise a non-protein coding gene e.g., sequences encoding antisense RNAs, RNAi, shRNAs and micro RNAs (miRNAs), miRNA sponges or decoys, recombinase delivery for conditional gene deletion, conditional (recombinase-dependent) expression, includes those required for the gene editing components described herein. The non-protein coding gene may also encode a tRNA, lRNA, tmRNA, piRNA, double stranded RNA, snRNA, snoRNA, and/or long non-coding RNA (lncRNA). In some cases, the non-protein coding gene can modulate the expression or the activity of a target gene or gene expression product. For example, the RNAs described herein may be used to inhibit gene expression in the CNS. In some cases, inhibition of gene expression refers to an inhibition by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In some cases, the protein product of the targeted gene may be inhibited by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. The gene can be either a wild type gene or a gene with at least one mutation. The targeted protein may be either a wild type protein or a protein with at least one mutation.

[0065] A therapeutic nucleic acid can modulate the expression or activity of a gene or gene expression product expressed from the gene that is implicated in a disease or disorder of the CNS. For example, the therapeutic nucleic

acid, in some cases is a gene or a modified version of the gene described herein. In some instances, the gene or gene expression product is inhibited. In some instances, the gene or gene expression product is enhanced.

[0066] In another example, the therapeutic nucleic acid comprises an effector gene expression product such as a gene editing component specific to target a gene therein. Non-limited examples of genes include target gene or gene expression product selected from ATP1A2, CACNA1A, SETD5, SHANK3, NF2, DNMT1, TCF4, RAI1, PEX1, ARSA, EIF2B5, EIF2B1, EIF2B2, NPC1, ADAR, MFSD8, STXBPI, PRICKLE2, PRRT2, IDUA, STX1B, Sarcoglycan Alpha (SGCA), glutamic acid decarboxylase 65 (GAD65), glutamic acid decarboxylase 67 (GAD67), CLN2, Nerve Growth Factor (NGF), glial cell derived neurotrophic factor (GDNF), Survival Of Motor Neuron 1, STXBP1, Telomeric (SMNI), Factor X (FIX), Retinoid Isomerohydrolase (RPE65), sarco/endoplasmic reticulum Ca²⁺-ATPase (SERCA2a), Glucocerebrosidase (GCase), galactocerebrosidase (GALC), CDKL5, Frataxin (FXN), Huntingtin (HTT), methyl-CpG binding protein 2 (MECP2), a peroxisomal biogenesis factor (PEX), progranulin (GRN), an anti-tubulin agent, copper-zinc superoxide dismutase (SOD1), iduronate 2 sulfatase (hIDS), Glucosylceramidase Beta (GBA), fragile X mental retardation 1 (FMR1), NPC Intracellular Cholesterol Transporter 1 (NPC1), SCN1A, C9orf72, NPS3 and a NLRP3 inflammasome. In some embodiments, the peroxisomal biogenesis factor (PEX) is selected from PEX1, PEX2, PEX3, PEX4, PEX5, PEX6, PEX7, PEX10, PEX11 β , PEX12, PEX13, PEX14, PEX16, PEX19, and PEX26. In some instances, the gene or gene expression product is inhibited. In some instances, the gene or gene expression product is enhanced.

AAV Vectors

[0067] Aspects disclosed herein comprise plasmid vectors comprising a nucleic acid sequence encoding the AAV capsids and AAV capsid proteins described herein. AAV vectors described herein are useful for the assembly of a rAAV and viral packaging of a heterologous nucleic acid. In addition, an AAV vector may encode a transgene comprising the heterologous nucleic acid.

[0068] An AAV vector can comprise a transgene, which in some cases encodes a heterologous gene expression product (e.g., therapeutic gene expression product, recombinant capsid protein, and the like). The transgene is in cis with two inverted terminal repeats (ITRs) flanking the transgene. The transgene may comprise a therapeutic nucleic acid encoding a therapeutic gene expression product. Due to the limited packaging capacity of the rAAV (~5 kB), in some cases, a longer transgene may be split between two AAV vectors, the first with 3' splice donor and the second with a 5' splice acceptor. Upon co-infection of a cell, concatemers form, which are spliced together to express a full-length transgene.

[0069] A transgene is generally inserted so that its expression is driven by the endogenous promoter at the integration site, namely the promoter that drives expression of the endogenous gene into which a transgene is inserted. In some instances, a transgene comprises a promoter and/or enhancer, for example a constitutive promoter or an inducible or tissue/cell specific promoter. As a non-limiting example, the promoter may be CMV promoter, a CMV- β -Actin-intron- β -Globin hybrid promoter (CAG), CBA promoter, FRDA or FXN promoter, UBC promoter, GUSB

promoter, NSE promoter, Synapsin promoter, MeCP2 promoter, GFAP promoter, H1 promoter, U6 promoter, NFL promoter, NFH promoter, SCN8A promoter, or PGK promoter. As a non-limiting example, promoters can be tissue-specific expression elements include, but are not limited to, human elongation factor 1 α -subunit (EF1 α), immediate-early cytomegalovirus (CMV), chicken β -actin (CBA) and its derivative CAG, the β glucuronidase (GUSB), and ubiquitin C (UBC). The transgene may include a tissue-specific expression elements for neurons such as, but not limited to, neuron-specific enolase (NSE), platelet-derived growth factor (PDGF), platelet-derived growth factor B-chain (PDGF- β), the synapsin (Syn), the methyl-CpG binding protein 2 (MeCP2), Ca²⁺/calmodulin-dependent protein kinase II (CaMKII), metabotropic glutamate receptor 2 (mGluR2), NFL, NFH, np32, PPE, Enk and EAAT2 promoters. The transgene may comprise a tissue-specific expression element for astrocytes such as, but not limited to, the glial fibrillary acidic protein (GFAP) and EAAT2 promoters. The transgene may comprise tissue-specific expression elements for oligodendrocytes such as, but not limited to, the myelin basic protein (MBP) promoter.

[0070] In some embodiments, the promoter is less than 1 kb. The promoter may have a length of 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350, 360, 370, 380, 390, 400, 410, 420, 430, 440, 450, 460, 470, 480, 490, 500, 510, 520, 530, 540, 550, 560, 570, 580, 590, 600, 610, 620, 630, 640, 650, 660, 670, 680, 690, 700, 710, 720, 730, 740, 750, 760, 770, 780, 790, 800 or more than 800. The promoter may have a length between 200-300, 200-400, 200-500, 200-600, 200-700, 200-800, 300-400, 300-500, 300-600, 300-700, 300-800, 400-500, 400-600, 400-700, 400-800, 500-600, 500-700, 500-800, 600-700, 600-800 or 700-800. The promoter may provide expression of the therapeutic gene expression product for a period of time in targeted tissues such as, but not limited to, the CNS. Expression of the therapeutic gene expression product may be for a period of 1 hour, 2, hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 9 hours, 10 hours, 11 hours, 12 hours, 13 hours, 14 hours, 15 hours, 16 hours, 17 hours, 18 hours, 19 hours, 20 hours, 21 hours, 22 hours, 23 hours, 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 8 days, 9 days, 10 days, 11 days, 12 days, 13 days, 2 weeks, 15 days, 16 days, 17 days, 18 days, 19 days, 20 days, 3 weeks, 22 days, 23 days, 24 days, 25 days, 26 days, 27 days, 28 days, 29 days, 30 days, 31 days, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 1 year, 13 months, 14 months, 15 months, 16 months, 17 months, 18 months, 19 months, 20 months, 21 months, 22 months, 23 months, 2 years, 3 years, 4 years, 5 years, 6 years, 7 years, 8 years, 9 years, 10 years, 11 years, 12 years, 13 years, 14 years, 15 years, 16 years, 17 years, 18 years, 19 years, 20 years, 21 years, 22 years, 23 years, 24 years, 25 years, 26 years, 27 years, 28 years, 29 years, 30 years, 31 years, 32 years, 33 years, 34 years, 35 years, 36 years, 37 years, 38 years, 39 years, 40 years, 41 years, 42 years, 43 years, 44 years, 45 years, 46 years, 47 years, 48 years, 49 years, 50 years, 55 years, 60 years, 65 years, or more than 65 years. Expression of the payload may be for 1-5 hours, 1-12 hours, 1-2 days, 1-5 days, 1-2 weeks, 1-3 weeks, 1-4 weeks, 1-2 months, 1-4 months, 1-6 months, 2-6 months, 3-6 months, 3-9 months, 4-8 months, 6-12 months, 1-2 years, 1-5 years, 2-5 years, 3-6 years, 3-8 years, 4-8 years or 5-10 years or 10-15 years, or 15-20 years, or

20-25 years, or 25-30 years, or 30-35 years, or 35-40 years, or 40-45 years, or 45-50 years, or 50-55 years, or 55-60 years, or 60-65 years.

[0071] An AAV vector can comprise a genome of a helper virus. Helper virus proteins are required for the assembly of a recombinant AAV (rAAV), and packaging of a transgene containing a heterologous nucleic acid into the rAAV. The helper virus genes are adenovirus genes E4, E2a and VA, that when expressed in the cell, assist with AAV replication. In some embodiments, an AAV vector comprises E2. In some embodiments, an AAV vector comprises E4. In some embodiments, an AAV vector comprises VA. In some instances, the AAV vector comprises one of helper virus proteins, or any combination.

[0072] The target gene or gene expression product for use in a transgene can be selected from ATP1A2, CACNA1A, SETD5, SHANK3, NF2, DNMT1, TCF4, RAI1, PEX1, ARSA, EIF2B5, EIF2B1, EIF2B2, NPC1, ADAR, MFSD8, STXBPI, PRICKLE2, PRRT2, IDUA, STX1B, Sarcoglycan Alpha (SGCA), glutamic acid decarboxylase 65 (GAD65), glutamic acid decarboxylase 67 (GAD67), CLN2, Nerve Growth Factor (NGF), glial cell derived neurotrophic factor (GDNF), Survival Of Motor Neuron 1, STXBP1, Telomeric (SMN1), Factor X (FIX), Retinoid Isomerohydrolase (RPE65), sarco/endoplasmic reticulum Ca2+-ATPase (SERCA2a), Glucocerebrosidase (GCase), galactocerebrosidase (GALC), CDKL5, Frataxin (FXN), Huntingtin (HTT), methyl-CpG binding protein 2 (MECP2), a peroxisomal biogenesis factor (PEX), progranulin (GRN), an anti-

tubulin agent, copper-zinc superoxide dismutase (SOD1), iduronate 2 sulfatase (hIDS), Glucosylceramidase Beta (GBA), fragile X mental retardation 1 (FMR1), NPC Intracellular Cholesterol Transporter 1 (NPC1), SCN1A, C9orf72, NPS3 and a NLRP3 inflammasome. In some embodiments, the peroxisomal biogenesis factor (PEX) is selected from PEX1, PEX2, PEX3, PEX4, PEX5, PEX6, PEX7, PEX10, PEX11B, PEX12, PEX13, PEX14, PEX16, PEX19, and PEX26.

[0073] An AAV vector can comprise a viral genome comprising a nucleic acid encoding the recombinant AAV (rAAV) capsid protein described herein. The viral genome can comprise a Replication (Rep) gene encoding a Rep protein, and Capsid (Cap) gene encoding an AAP protein in the first open reading frame (ORF1) or a Cap protein in the second open reading frame (ORF2). The Rep protein is selected from Rep78, Rep68, Rep52, and Rep40. In some instances, the Cap gene is modified encoding a modified AAV capsid protein described herein. A wild-type Cap gene encodes three proteins, VP1, VP2, and VP3. In some cases, VP1 is modified. In some cases, VP2 is modified. In some cases, VP3 is modified. In some cases, all three VP1-VP3 are modified. The AAV vector can comprise nucleic acids encoding wild-type Rep78, Rep68, Rep52, Rep40 and AAP proteins.

[0074] In some instances, the AAV9 VP1 gene provided in SEQ ID NO: 86 shown in Table 3, is modified to include any one of SEQ ID NOS: 67-85 in FIG. 1. The AAV vector described herein may be used to produce a variant AAV capsid by the methods described herein.

TABLE 3

VP1 Capsid Protein Nucleic Acid Sequences		
SEQ ID NO:	Identifier	Sequence
86	AAV9 > AY530579.1 Adeno-associated virus 9 isolate hu. 14 capsid protein VP1 (cap) gene, complete cds	ATGGCTGCCGATGGTTATCTTCCAGAT TGGCTCGAGGACACCTTAGTGAAGG AATTCGCGAGTGGTGGCTTTGAAACC TGGAGCCCCCAACCAAGGCAATC AACAACTCAAGACAACGCTCGAGGT CTTGTGCTTCCGGGTTACAATACCTT GGACCCGGCAACGACTCGACAAGGG GGAGCCGGTCAACGACGACGCGCG CGGCCCTCGAGCAGCACAAGCCTAC GACCAGCAGCTCAAGGCCGAGACAA CCCGTACCTCAAGTACAACACGCGCA CGCCGAGTTCCAGGAGCGGCTCAAAG AAGATACGTCTTTTGGGGGCAACCTCG GGCGAGCAGTCTTCCAGGCCAAAAG AGGCTTCTTGAACCTCTTGGTCTGGTT GAGGAAGCGGCTAAGACGGCTCCTGG AAAGAAGAGGCTGTAGAGCAGTCTC CTCAGGAACCGGACTCCTCCGCGGTA TTGGCAAATCGGGTGCACAGCCCGCTA AAAAGAGACTCAATTTGGGTCAGACTG GCGACACAGAGTCAGTCCCAGACCCTC AACCAATCGGAGAACCTCCCGCAGCC CCCTCAGGTGTGGGATCTTTACAATG GCTTCAGGTGGTGGCGACCAAGTGGCA GACAATAACGAAGGTGCCGATGGAGT GGGTAGTTCTCGGGAATTTGGCATTG CGATTCCCAATGGCTGGGGGACAGAG TCATACCCACCAGCACCCGAACCTGGG CCCTGCCCACTACAACAATCACCTCT ACAAGCAAATCTCCAACAGCACATCTG GAGGATCTTCAAATGACAACGCTACT TCGGCTACAGCACCCCTGGGGGTATT TTGACTTCAACAGATTCCACTGCCACT TCTCACCAGTGACTGGCAGCGACTCA TCAACAACAAGTGGGATTCCGCGCTA

TABLE 3-continued

VP1 Capsid Protein Nucleic Acid Sequences		
SEQ ID NO:	Identifier	Sequence
		AGCGACTCAACTTCAAGCTCTTCAACA TTCAGGTCAAAGAGGTTACGGACAAC AATGGAGTCAAGACCATCGCCAATAA CCTTACCAGCACGGTCCAGGTCTTCAC GGACTCAGACTATCAGTCCCGTACGT GCTCGGGTCGGCTCAGAGGGCTGCCT CCCGCCGTCCAGCGGACGTTTTTCAT GATTCCCTCAGTACGGGTATCTGACGCT TAATGATGGAAGCCAGGCCGTGGGTC GTTCTGTCCTTTACTGCCTGGAATATT CCCGTCGCAAATGCTAAGAACGGGTA ACAACTTCCAGTTCAGCTACGAGTTTG AGAACGTACCTTTCATAGCAGCTACG CTCACAGCCAAAGCCTGGACCGACTA ATGAATCCACTCATCGACCAATACTTG TACTATCTCTCAAAGACTATTAACGGT TCTGGACAGAATCAACAAACGCTAAA ATTCAGTGTGGCCGGACCCAGCAACAT GGCTGTCCAGGGAAGAACTACATAC CTGGACCCAGCTACCGACAACAACGT GTCTCAACCACTGTGACTCAAAACAAC AACAGCGAATTGCTTGGCCTGGAGCT TCTTCTTGGGCTCTCAATGGACGTAAT AGCTTGATGAATCCTGGACCTGCTATG GCCAGCCACAAGAAGGAGAGGACCG TTTCTTTCCTTTGCTGGATCTTTAATT TTTGGCAAACAAGGAAGTGAAGAGA CAACGTGGATGCGGACAAAGTCATGA TAACCAACGAAGAAGAAATTTAAACT ACTAACCCGGTAGCAACGGAGTCTTAT GGACAAGTGGCCACAACACCAGAG TGCCCAAGCACAGGCGCAGACCGGCT GGGTTCAAAACCAAGGAATACTTCCG GGTATGGTTTGGCAGGACAGAGATGT GTACCTGCAAGGACCCATTTGGGCCAA AATTCTTCACACGGACGGCAACTTTCA CCCTTCTCCGTGATGGGAGGGTTGG AATGAAGCACCCGCTCCTCAGATCCT CATCAAAACACACCTGTACCTGCGGA TCCTCCAACGGCCTTCAACAAGGACAA GCTGAATCTTTTCATACCCAGTATTC TACTGGCCAAGTCAGCGTGGAGATCG AGTGGGAGCTGCAGAAGGAAACAGC AAGCGCTGGAACCCGGAGATCCAGTA CACTTCCAATATTACAAGTCTAATAA TGTTGAATTTGCTGTTAATACTGAAGG TGTATATAGTGAACCCCGCCCATTTGG CACCAGATACCTGACTCGTAATCTGTA A

Methods of Producing rAAVs

[0075] Disclosed herein are methods of producing the AAV capsids comprising the AAV capsid proteins and viral vector encoding a therapeutic nucleic acid. The AAV capsid proteins are produced by introducing into a cell (e.g., immortalized stem cell) a first vector containing a transgene cassette flanked by inverted terminal repeat (ITR) sequences from a parental AAV virus (the transgene cassette has a promoter sequence that drives transcription of a heterologous nucleic acid in the nucleus of the target cell), a second vector encoding the AAV genome with a AAV capsid protein (encoding the AAV Rep gene as well as the modified Cap gene for the variant being produced), and a third vector encoding helper virus proteins, required for assembly of the AAV capsid structure and packaging of the transgene in the modified AAV capsid structure. The assembled AAV capsid can be isolated and purified from the cell using suitable

methods known in the art. Table 1 provide DNA sequences for using in the methods described herein.

[0076] The transgenes contained in a recombinant AAV (rAAV) vector and encapsidated by the AAV capsid proteins of the present disclosure are also provided herein. The transgenes disclosed herein are delivered to a subject for a variety of purposes, such as to treat a disease or condition in the subject. The transgene can be gene editing components that modulate the activity or expression of a target gene or gene expression product. Alternatively, the transgene is a gene encoding a therapeutic gene expression product that is effective to modulate the activity or expression of itself, or another target gene or gene expression product.

[0077] Aspects disclosed herein provide methods of manufacturing rAAV virus or virus particles comprising: (a) introducing into a cell a nucleic acid comprising: (i) first vector containing a transgene cassette flanked by inverted terminal repeat (ITR) sequences from a parental AAV virus

(the transgene cassette has a promoter sequence that drives transcription of a heterologous nucleic acid in the nucleus of the target cell); (ii) a second vector encoding the AAV genome with a AAV capsid protein of the present invention; and (iii) a vector encoding helper virus proteins, required for assembly of the AAV capsid structure and packaging of the transgene in the modified AAV capsid structure; (b) expressing in the cell the AAV capsid protein described herein; (c) assembling an AAV particle comprising the AAV capsid proteins disclosed herein; and (d) packaging the AAV particle. In some instances, the cell is mammalian. In some instances, the cell is immortalized. In some instances, the immortalized cell is an embryonic stem cell. In some instances, the embryonic stem cell is a human embryonic stem cell. In some instances, the human embryonic stem cell is a human embryonic kidney 293 (HEK-293) cell. In some instances, the Cap gene is derived from the deoxyribose nucleic acid (DNA) provided in SEQ ID NO: 86. In some instances, the 5' ITR and the 3' ITR are derived from an AAV2 serotype. In some instances, the 5' ITR and the 3' ITR are derived from an AAV5 serotype. In some instances, the 5' ITR and the 3' ITR are derived from an AAV9 serotype. In some instances, the first nucleic acid sequence and the second nucleic acid sequence are in trans. In some instances, the first nucleic acid sequence and the second nucleic acid sequence are in cis. In some instances, the first nucleic acid sequence, the second nucleic acid sequence and the third nucleic acid sequence, are in trans.

[0078] The Cap gene disclosed here comprises any one of SEQ ID NOS: 67-85 from FIG. 1, which are DNA sequences encoding the modified AAV capsid protein portions of the present disclosure.

[0079] In some instances, the methods comprise packing the first nucleic acid sequence encoding the therapeutic gene expression product such that it becomes encapsidated by the modified AAV capsid protein. In some embodiments, the rAAV particles are isolated, concentrated, and purified using suitable viral purification methods, such as those described herein.

[0080] In some cases, rAAVs of the present disclosure are generated using the methods described in Challis, R. C. et al. Nat. Protoc. 14, 379 (2019). Briefly, triple transfection of HEK293T cells (ATCC) using polyethylenimine (PEI) is performed, viruses are collected after 120 hours from both cell lysates and media and purified over iodixanol. In a non-limiting example, the rAAVs are generated by triple transfection of precursor cells (e.g., HEK293T) cells using a standard transfection protocol (e.g., PEI). Viral particles are harvested from the media after a period of time (e.g., 72 h post transfection) and from the cells and media at a later point in time (e.g., 120 h post transfection). Virus present in the media is concentrated by precipitation with 8% polyethylene glycol (PEG) and 500 mM sodium chloride and the precipitated virus is added to the lysates prepared from the collected cells. The viruses are purified over iodixanol (Optiprep, Sigma) step gradients (15%, 25%, 40% and 60%). Viruses are concentrated and formulated in PBS. Virus titers are determined by measuring the number of DNaseI-resistant vector genome copies (VGs) using qPCR and the linearized genome plasmid as a control.

[0081] The cell can be selected from a human, a primate, a murine, a feline, a canine, a porcine, an ovine, a bovine, an equine, an epine, a caprine and a lupine host cell. In some instances, the cell is a progenitor or precursor cell, such as

a stem cell. In some instances, the stem cell is a mesenchymal cell, embryonic stem cell, induced pluripotent stem cell (iPSC), fibroblast or other tissue specific stem cell. The cell can be immortalized. In some cases, the immortalized cell is a HEK293cell. In some instances, the cell is a differentiated cell. Based on the disclosure provided, it is expected that this system can be used in conjunction with any transgenic line expressing a recombinase in the target cell type of interest to develop AAV capsids that more efficiently transduce that target cell population.

Methods of Treatment

[0082] Disclosed herein are methods of treating a disease or condition, or a symptom of the disease or condition, in a subject, comprising administering of therapeutically effective amount of one or more compositions (e.g., rAAV particle, AAV vector, pharmaceutical composition) disclosed herein to the subject. In some embodiments, the composition is a rAAV capsid protein described herein. In some embodiments, the composition is an isolated and purified rAAV capsid protein described herein. In some embodiments, the rAAV particle encapsidates an AAV vector comprising a transgene (e.g., therapeutic nucleic acid). In some embodiments, the composition is a rAAV capsid protein described herein conjugated with a therapeutic agent disclosed herein. In some embodiments, the composition is a pharmaceutical composition comprising the rAAV particle and a pharmaceutically acceptable carrier. In some embodiments, the one or more compositions are administered to the subject alone (e.g., stand-alone therapy). In some embodiments, the composition is a first-line therapy for the disease or condition. In some embodiments, the composition is a second-line, third-line, or fourth-line therapy, for the disease or condition.

[0083] Recombinant adeno-associated virus (rAAV) mediated gene delivery leverages the AAV mechanism of viral transduction for nuclear expression of an episomal heterologous nucleic acid (e.g., a transgene, therapeutic nucleic acid). Upon delivery to a host in vivo environment, a rAAV will (1) bind or attach to cellular surface receptors on the target cell, (2) endocytose, (3) traffic to the nucleus, (4) uncoat the virus to release the encapsidated heterologous nucleic acid, (5) convert of the heterologous nucleic acid from single-stranded to double-stranded DNA as a template for transcription in the nucleus, and (6) transcribe of the episomal heterologous nucleic acid in the nucleus of the host cell ("transduction"). rAAVs engineered to have an increased transduction enrichment (transcription of the episomal heterologous nucleic acid in the host cell) are desirable for gene therapy applications.

[0084] Aspects disclosed herein provide methods of treating a disease or condition in a subject, the method comprising administering to the subject a therapeutically effective amount of the rAAV of the present disclosure, or the pharmaceutical formulation of the present disclosure, wherein the gene product is a therapeutic gene product. In some embodiments, the administering is by intracranial, intraventricular, intracerebroventricular, intravenous, intraarterial, intranasal, intrathecal, intracisternae magna, or subcutaneous.

[0085] Provided here, are methods of treating a disease or a condition associated with an aberrant expression or activity of a target gene or gene expression product thereof, the method comprising modulating the expression or the activity

of a target gene or gene expression product in a subject by administering a rAAV encapsidating a heterologous nucleic acid of the present disclosure. In some instances, the expression or the activity of the target gene or gene expression product is decreased, relative to that in a normal (non-diseased) individual; and administering the rAAV to the subject is sufficient to increase the expression of the activity of the target gene or gene expression product. In some instances, the expression or the activity of the gene or gene expression product is increased, relative to that in a normal individual; and administering the rAAV to the subject is sufficient to decrease the expression or the activity of the target gene or gene expression product. In a non-limiting example, a subject diagnosed with Alzheimer's disease, which is caused, in some cases, by a gain-of-function of a Presenilin 1 and/or Presenilin 2 (encoded by the gene PSEN1 and PSEN2, respectively) is administered a rAAV disclosed herein encapsidating a therapeutic nucleic acid that is a silencing RNA (siRNA), or other RNAi with a loss-of-function effect on PSEN1 mRNA.

[0086] Also provided are methods of preventing a disease or condition disclosed herein in a subject comprising administering to the subject a therapeutically effective amount of an rAAV vector comprising a nucleic acid sequence encoding a therapeutic gene expression product described herein. The rAAV vector may be encapsidated in the modified capsid protein or rAAV viral particle described herein. In some instances, the therapeutic gene expression product is effective to modulate the activity or expression of a target gene or gene expression product.

[0087] Disclosed herein are methods of treating a disease or condition in a subject by administering a composition comprising a rAAV disclosed herein. An advantage of the rAAVs disclosed herein, is that the rAAV may be used to treat virtually any disease or condition that would benefit from a transgene therapy, including but not limited to spinal muscular atrophy (SMA), amyotrophic lateral sclerosis (ALS), Parkinson's disease, Pompe disease, mucopolysaccharidosis type II, fragile X syndrome, STXBPI encephalopathy, Krabbe disease, Huntington's disease, Alzheimer's disease, Batten disease, lysosomal storage disorders, glioblastoma multiforme, Rett syndrome, Leber's congenital amaurosis, Late infantile neuronal ceroid lipofuscinosis (LINCL), chronic pain, stroke, spinal cord injury, traumatic brain injury and lysosomal storage disorders.

[0088] In some cases, the disease or condition is localized to a particular in vivo environment in the subject, e.g., the CNS. The compositions of the present disclosure are particularly useful for the treatment of the diseases or conditions described herein because they specifically or more efficiently target the in vivo environment and deliver a therapeutic nucleic acid engineered to modulate the activity or the expression of a target gene expression product involved with the pathogenesis or pathology of the disease or condition.

[0089] Provided herein are methods of treating a disease or a condition, or a symptom of the disease or condition, in a subject, comprising: (a) diagnosing a subject with a disease or a condition affecting a target in vivo environment; and (b) treating the disease or the condition by administering to the subject a therapeutically effective amount of a composition disclosed herein (e.g., rAAV particle, AAV vector, pharmaceutical composition), wherein the composition is engineered with an increased expression.

[0090] Disclosed herein are methods of treating a disease or a condition, or a symptom of the disease or the condition, afflicting a target in a subject comprising: (a) administering to the subject a composition (e.g., rAAV particle, AAV vector, pharmaceutical composition); and (b) expressing the therapeutic nucleic acid into a target in vivo environment in the subject with an increased transduction enrichment.

[0091] In some embodiments, methods further comprise reducing or ablating delivery of the heterologous nucleic acid in an off-target in vivo environment, such as the liver. In some embodiments, delivery is characterized by an increase in enrichment of transduction (e.g., of the heterologous nucleic acid) in the CNS.

[0092] In some embodiments, methods of treating a disease or condition affecting the CNS comprise administering a rAAV particle to a CNS in a subject, the rAAV particle comprising an rAAV capsid protein comprising an insertion of about, five, six, or seven amino acids of an amino acid sequence provided in Table 1 and FIG. 1, at an amino acid position 588-589 in a parental AAV capsid protein. In some embodiments, methods of treating a disease or condition affecting the CNS comprise administering a rAAV particle to a CNS in a subject, the rAAV particle comprising an rAAV capsid protein comprising an insertion of about, five, six, or seven amino acids of an amino acid sequence as well as one or more substitution at amino acid found at amino acid positions 587-590 [AQAQ] such as provided in Table 1 and FIG. 1. In some embodiments, the parental AAV capsid protein is AAV9 capsid protein (for e.g., provided in SEQ ID NO: 1).

[0093] Also provided are methods of modulating a target gene expression product, the methods comprising administering to a subject in need thereof a composition (e.g., rAAV particle, AAV vector, pharmaceutical composition) disclosed herein. For example, methods provided herein comprise administering to a subject a rAAV with a rAAV capsid protein encapsidating a viral vector comprising a heterologous nucleic acid that modulates the expression or the activity of the target gene expression product.

[0094] The term "normal individual" refers to an individual that is not afflicted with the disease or the condition characterized by the variation in expression or activity of the gene or gene expression product thereof.

[0095] In some embodiments, the disease or condition of the CNS is selected from Absence of the Septum Pellucidum, Acid Lipase Disease, Acid Maltase Deficiency, Acquired Epileptiform Aphasia, Acute Disseminated Encephalomyelitis, Attention Deficit-Hyperactivity Disorder (ADHD), Adie's Pupil, Adie's Syndrome, Adrenoleukodystrophy, Agenesis of the Corpus Callosum, Agnosia, Aicardi Syndrome, Aicardi-Goutieres Syndrome Disorder, AIDS-Neurological Complications, Alexander Disease, Alpers' Disease, Alternating Hemiplegia, Alzheimer's Disease, Amyotrophic Lateral Sclerosis (ALS), Anencephaly, Aneurysm, Angelman Syndrome, Angiomas, Anoxia, Antiphospholipid Syndrome, Aphasia, Apraxia, Arachnoid Cysts, Arachnoiditis, Arnold-Chiari Malformation, Arteriovenous Malformation, Asperger Syndrome, Ataxia, Ataxia Telangiectasia, Ataxias and Cerebellar or Spinocerebellar Degeneration, Atrial Fibrillation and Stroke, Attention Deficit-Hyperactivity Disorder, Autism Spectrum Disorder, Autonomic Dysfunction, Back Pain, Barth Syndrome, Batten Disease, Becker's Myotonia, Behcet's Disease, Bell's Palsy, Benign Essential Blepharospasm, Benign Focal

Amyotrophy, Benign Intracranial Hypertension, Bernhardt-Roth Syndrome, Binswanger's Disease, Blepharospasm, Bloch-Sulzberger Syndrome, Brachial Plexus Birth Injuries, Brachial Plexus Injuries, Bradbury-Eggleston Syndrome, Brain and Spinal Tumors, Brain Aneurysm, Brain Injury, Brown-Sequard Syndrome, Bulbospinal Muscular Atrophy, Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy (CADASIL), Canavan Disease, Carpal Tunnel Syndrome, Causalgia, Cavernomas, Cavernous Angioma, Cavernous Malformation, Central Cervical Cord Syndrome, Central Cord Syndrome, Central Pain Syndrome, Central Pontine Myelinolysis, Cephalic Disorders, Ceramidase Deficiency, Cerebellar Degeneration, Cerebellar Hypoplasia, Cerebral Aneurysms, Cerebral Arteriosclerosis, Cerebral Atrophy, Cerebral Biberi, Cerebral Cavernous Malformation, Cerebral Gigantism, Cerebral Hypoxia, Cerebral Palsy, Cerebro-Oculo-Facio-Skeletal Syndrome (COFS), Charcot-Marie-Tooth Disease, Charcot-Marie-Tooth syndrome, classical rhizomelic chondrodysplasia punctata (RCDP), Chiari Malformation, Cholesterol Ester Storage Disease, Chorea, Choreoacanthocytosis, Chronic Inflammatory Demyelinating Polyneuropathy (CIDP), Chronic Orthostatic Intolerance, Chronic Pain, Cockayne Syndrome, Cockayne Syndrome Type II, Coffin Lowry Syndrome, Colpocephaly, Coma, Complex Regional Pain Syndrome, Congenital Facial Diplegia, Congenital Myasthenia, Congenital Myopathy, Congenital Vascular Cavernous Malformations, Corticobasal Degeneration, Cranial Arteritis, Craniosynostosis, Cree encephalitis, Creutzfeldt-Jakob Disease, Cumulative Trauma Disorders, Cushing's Syndrome, Cytomegalic Inclusion Body Disease, Cytomegalovirus Infection, Dancing Eyes-Dancing Feet Syndrome, Dandy-Walker Syndrome, Dawson Disease, Deafness, De Morsier's Syndrome, Dejerine-Klumpke Palsy, Dementia, Dementia-Multi-Infarct, Dementia-Semantic, Dementia-Subcortical, Dementia With Lewy Bodies, Dentate Cerebellar Ataxia, Dentatorubral Atrophy, Dermatomyositis, Developmental Dyspraxia, Devic's Syndrome, Diabetic Neuropathy, Diffuse Sclerosis, Dravet Syndrome, Duchenne muscular dystrophy, Dysautonomia, Dysgraphia, Dyslexia, Dysphagia, Dyspraxia, Dyssynergia Cerebellaris Myoclonica, Dyssynergia Cerebellaris Progressiva, Dystonias, Early Infantile Epileptic Encephalopathy, Empty Sella Syndrome, Encephalitis, Encephalitis Lethargica, Encephaloceles, Encephalopathy, Encephalopathy (familial infantile), Encephalotrigeminal Angiomatosis, Epilepsy, Epileptic Hemiplegia, Erb's Palsy, Erb-Duchenne and Dejerine-Klumpke Palsies, Essential Tremor, Extrapontine Myelinolysis, Fabry Disease, Fahr's Syndrome, Fainting, Familial Dysautonomia, Familial Hemangioma, Familial Idiopathic Basal Ganglia Calcification, Familial Periodic Paralysis, Familial Spastic Paralysis, Farber's Disease, Febrile Seizures, Fibromuscular Dysplasia, Fisher Syndrome, Floppy Infant Syndrome, Foot Drop, Fragile X syndrome, Friedreich's Ataxia, Frontotemporal Dementia (FTD), Gaucher Disease, Generalized Gangliosidosis, Gerstmann's Syndrome, Gerstmann-Straussler-Scheinker Disease, Giant Axonal Neuropathy, Giant Cell Arteritis, Giant Cell Inclusion Disease, glioblastoma, Globoid Cell Leukodystrophy, Glossopharyngeal Neuralgia, Glycogen Storage Disease, Guillain-Barre Syndrome, Hallervorden-Spatz Disease, Head Injury, Headache, Hemispheric Continua, Hemifacial Spasm, Hemiplegia Alterans, Hereditary Neuropathies, Hereditary Spastic Paraplegia,

Heredopathia Atactica Polyneuritiformis, Herpes Zoster, Herpes Zoster Oticus, Hirayama Syndrome, Holmes-Adie syndrome, Holoprosencephaly, HTLV-1 Associated Myelopathy, Hughes Syndrome, Huntington's Disease, Hydranencephaly, Hydrocephalus, Hydrocephalus-Normal Pressure, Hydromyelia, Hypercortisolism, Hypersomnia, Hypertonia, Hypotonia, Hypoxia, Immune-Mediated Encephalomyelitis, Inclusion Body Myositis, Incontinentia Pigmenti, Infantile Hypotonia, Infantile Neuroaxonal Dystrophy, Infantile Phytanic Acid Storage Disease, Infantile Refsum Disease (IRD), Infantile Spasms, Inflammatory Myopathies, Iniencephaly, Intestinal Lipodystrophy, Intracranial Cysts, Intracranial Hypertension, Isaacs' Syndrome, Joubert Syndrome, Kearns-Sayre Syndrome, Kennedy's Disease, Kinsbourne syndrome, Kleine-Levin Syndrome, Klippel-Feil Syndrome, Klippel-Trenaunay Syndrome (KTS), Kliever-Bucy Syndrome, Korsakoff's Amnesic Syndrome, Krabbe Disease, Kugelberg-Welander Disease, Kuru, Lambert-Eaton Myasthenic Syndrome, Landau-Kleffner Syndrome, Lateral Femoral Cutaneous Nerve Entrapment, Lateral Medullary Syndrome, Learning Disabilities, Leigh's Disease, Lennox-Gastaut Syndrome, Lesch-Nyhan Syndrome, Leukodystrophy, Levine-Critchley Syndrome, Lewy Body Dementia, Lipid Storage Diseases, Lipoid Proteinosis, Lissencephaly, Locked-In Syndrome, Lou Gehrig's Disease, Lupus-Neurological Sequelae, Lyme Disease-Neurological Complications, Machado-Joseph Disease, Macrocephaly, Maple syrup urine disease, Megalencephaly, Melkersson-Rosenthal Syndrome, Meningitis, Meningitis and Encephalitis, Menkes Disease, Menkes syndrome, Meralgia Paresthetica, Metachromatic Leukodystrophy, Microcephaly, Migraine, Miller Fisher Syndrome, Mini Stroke, Mitochondrial Myopathy, Moebius Syndrome, Monomelic Amyotrophy, Motor Neuron Diseases, Moyamoya Disease, Mucopolysaccharidosis, Mucopolysaccharidosis II, Multi-Infarct Dementia, Multifocal Motor Neuropathy, Multiple Sclerosis, Multiple System Atrophy, Multiple System Atrophy with Orthostatic Hypotension, Muscular Dystrophy, Myasthenia-Congenital, Myasthenia Gravis, Myelinoclastic Diffuse Sclerosis, Myoclonic Encephalopathy of Infants, Myoclonus, Myopathy, Myopathy-Congenital, Myopathy-Thyrotoxic, Myotonia, Myotonia Congenita, Myotonic dystrophy, Narcolepsy, Neuroacanthocytosis, Neurodegeneration with Brain Iron Accumulation, Neurofibromatosis, Neuroleptic Malignant Syndrome, Neurological Complications of AIDS, Neurological Complications of Lyme Disease, Neurological Consequences of Cytomegalovirus Infection, Neurological Manifestations of Pompe Disease, Neurological Sequelae Of Lupus, Neuromyelitis Optica, Neuromyotonia, Neuronal Ceroid Lipofuscinosis, Neuronal Migration Disorders, Neuropathy-Hereditary, Neurosarcoidosis, Neurosyphilis, Neurotoxicity, Nevus Cavernosus, Niemann-Pick Disease, O'Sullivan-McLeod Syndrome, Occipital Neuralgia, Ohtahara Syndrome, Olivopontocerebellar Atrophy, Opsoclonus Myoclonus, Orthostatic Hypotension, Overuse Syndrome, Pain-Chronic, Pantothenate Kinase-Associated Neurodegeneration, Paraneoplastic Syndromes, Paresthesia, Parkinson's Disease, Paroxysmal Choreoathetosis, Paroxysmal Hemispheric, Parry-Romberg, Pelizaeus-Merzbacher Disease, Pena Shokeir II Syndrome, Perineural Cysts, Periodic Paralysis, Peripheral Neuropathy, Periventricular Leukomalacia, Persistent Vegetative State, Pervasive Developmental Disorders, Phenylketonuria, Phytanic Acid Storage

Disease, Pick's Disease, Pinched Nerve, *Piriformis* Syndrome, Pituitary Tumors, Polymyositis, Pompe Disease, Porencephaly, Post-Polio Syndrome, Postherpetic Neuralgia, Postinfectious Encephalomyelitis, Postural Hypotension, Postural Orthostatic Tachycardia Syndrome, Postural Tachycardia Syndrome, Prader-Willi syndrome, Primary Dentatum Atrophy, Primary Lateral Sclerosis, Primary Progressive Aphasia, Prion Diseases, Progressive Hemifacial Atrophy, Progressive Locomotor Ataxia, Progressive Multifocal Leukoencephalopathy, Progressive Sclerosing Poliodystrophy, Progressive Supranuclear Palsy, Prosopagnosia, Pseudo-Torch syndrome, Pseudotuberculosis syndrome, Pseudotumor Cerebri, Psychogenic Movement, Ramsay Hunt Syndrome I, Ramsay Hunt Syndrome II, Rasmussen's Encephalitis, Reflex Sympathetic Dystrophy Syndrome, Refsum Disease, Refsum Disease-Infantile, Repetitive Motion Disorders, Repetitive Stress Injuries, Restless Legs Syndrome, Retrovirus-Associated Myelopathy, Rett Syndrome, Reye's Syndrome, Rheumatic Encephalitis, Riley-Day Syndrome, Sacral Nerve Root Cysts, Saint Vitus Dance, Salivary Gland Disease, Sandhoff Disease, Schilder's Disease, Schizencephaly, Seitelberger Disease, Seizure Disorder, Semantic Dementia, Septo-Optic Dysplasia, Severe Myoclonic Epilepsy of Infancy (SMEI), Shaken Baby Syndrome, Shingles, Shy-Drager Syndrome, Sjogren's Syndrome, Sleep Apnea, Sleeping Sickness, Sotos Syndrome, Spasticity, Spina *Bifida*, Spinal Cord Infarction, Spinal Cord Injury, Spinal Cord Tumors, Spinal Muscular Atrophy, Spinocerebellar ataxia, Spinocerebellar Atrophy, Spinocerebellar Degeneration, Steele-Richardson-Olszewski Syndrome, Stiff-Person Syndrome, Striatonigral Degeneration, Stroke, Sturge-Weber Syndrome, STXBPI encephalopathy, Subacute Sclerosing Panencephalitis, Subcortical Arteriosclerotic Encephalopathy, Short-lasting, Unilateral, Neuralgiform (SUNCT) Headache, Swallowing Disorders, Sydenham Chorea, Syncope, Syphilitic Spinal Sclerosis, Syringohydromyelia, Syringomyelia, Systemic Lupus Erythematosus, Tabes *Dorsalis*, Tangier disease, Tardive Dyskinesia, Tarlov Cysts, Tay-Sachs Disease, Temporal Arteritis, Tethered Spinal Cord Syndrome, Thomsen's Myotonia, Thoracic Outlet Syndrome, Thyrotoxic Myopathy, Tic Douloureux, Todd's Paralysis, Tourette Syndrome, Transient Ischemic Attack, Transmissible Spongiform Encephalopathies, Transverse Myelitis, Traumatic Brain Injury, Tremor, Trigeminal Neuralgia, Tropical Spastic Paraparesis, Troyer Syndrome, Tuberos Sclerosis, Vascular Erectile Tumor, Vasculitis Syndromes of the Central Nervous Systems, Von Economo's Disease, Von Hippel-Lindau Disease (VHL), Von Hippel-Lindau syndrome, Von Recklinghausen's Disease, Wallenberg's Syndrome, Werdnig-Hoffman Disease, Wernicke-Korsakoff Syndrome, West Syndrome, Whiplash, Whipple's Disease, Williams Syndrome, Wilson Disease, Wolman's Disease, X-Linked Spinal and Bulbar Muscular Atrophy and Zellweger syndrome.

[0096] In some embodiments, the pharmaceutical formulation comprises a therapeutic nucleic acid encoding a therapeutic gene expression product. In some instances, the therapeutic gene expression product is effective to modulate an activity or an expression of a target gene or gene expression product selected from ATP1A2, CACNA1A, SETD5, SHANK3, NF2, DNMT1, TCF4, RAI1, PEX1, ARSA, EIF2B5, EIF2B1, EIF2B2, NPC1, ADAR, MFSD8, STXBPI, PRICKLE2, PRRT2, IDUA, STX1B, Sarcoglycan Alpha (SGCA), glutamic acid decarboxylase 65 (GAD65),

glutamic acid decarboxylase 67 (GAD67), CLN2, Nerve Growth Factor (NGF), glial cell derived neurotrophic factor (GDNF), Survival Of Motor Neuron 1, STXBPI, Telomeric (SMNI), Factor X (FIX), Retinoid Isomerohydrolase (RPE65), sarco/endoplasmic reticulum Ca²⁺-ATPase (SERCA2a), Glucocerebrosidase (GCase), galactocerebrosidase (GALC), CDKL5, Frataxin (FXN), Huntingtin (HTT), methyl-CpG binding protein 2 (MECP2), a peroxisomal biogenesis factor (PEX), progranulin (GRN), an anti-tubulin agent, copper-zinc superoxide dismutase (SOD1), iduronate 2 sulfatase (hIDS), Glucosylceramidase Beta (GBA), fragile X mental retardation 1 (FMR1), NPC Intracellular Cholesterol Transporter 1 (NPC1), SCN1A, C9orf72, NPS3 and a NLRP3 inflammasome. In some embodiments, the peroxisomal biogenesis factor (PEX) is selected from PEX1, PEX2, PEX3, PEX4, PEX5, PEX6, PEX7, PEX10, PEX11B, PEX12, PEX13, PEX14, PEX16, PEX19, and PEX26.

[0097] In some aspects, other examples of genes involved in CNS diseases or disorders include MAPT, IDUA, SNCA, ATXN2, Ube3a, GNS, HGSNAT, NAGLU, SGSH, CLN1, CLN3, CLN4, CLN5, CLN6, CLN7, CLN8, CTSD, ABCD1, HEXA, HEXB, ASM, ASPA, GLB1, AADC, MFN2, GNAO1, SYNGAP1, GRIN2A, GRIN2B, KCNQ2, EPM2A, NHLRC1, SLC6A1, SLC13A5, SURF1, GBE1, ATXN1, ATXN3, and ATXN7.

[0098] In some instances, the therapeutic gene expression product comprises gene editing components. In some instances, the gene editing components are selected from an artificial site-specific RNA endonuclease (ASRE), a zinc finger endonuclease (ZFN), a transcription factor like effector nuclease (TALEN), a clustered regularly interspaced short palindromic repeats (CRISPR)/Cas enzyme, and a CRISPR/Cas guide RNA.

[0099] In some instances, the expression of a gene or expression or activity of a gene expression product is inhibited by the administration of the composition to the subject. In some instances, the expression of a gene or the expression or the activity of a gene expression product is enhanced by the administration of the composition to the subject.

Formulations, Dosages, and Routes of Administration

[0100] Disclosed herein are methods comprising delivering a rAAV particle encapsidating a heterologous nucleic acid to the CNS in a subject, the rAAV particle comprising (i) an increased transduction of the heterologous nucleic acid in the CNS, wherein the rAAV particle has an rAAV capsid protein comprising an insertion of five, six, or seven amino acids of an amino acid sequence provided in Table 1 and FIG. 1, at an amino acid position 588-589 in a parental AAV capsid protein or comprising an rAAV capsid protein comprising an insertion of about, five, six, or seven amino acids of an amino acid sequence as well as one or more substitution at amino acid found at amino acid positions 587-590 [AQAA] such as provided in Table 1 and FIG. 1.

[0101] In general, methods disclosed herein comprise administering a therapeutic rAAV composition by systemic administration. In some instances, methods comprise administering a therapeutic rAAV composition by intravenous ("i.v.") administration. One may administer therapeutic rAAV compositions by additional routes, such as subcutaneous injection, intramuscular injection, intradermal injection,

tion, transdermal injection, percutaneous administration, intranasal administration, intralymphatic injection, rectal administration intragastric administration, intraocular administration, intracerebroventricular administration, intrathecally, intracisternal, or any other suitable parenteral administration. Routes, dosage, time points, and duration of administering therapeutics may be adjusted. In some embodiments, administration of therapeutics is prior to, or after, onset of either, or both, acute and chronic symptoms of the disease or condition. Other routes of delivery to the CNS include, but are not limited to intracranial administration, lateral cerebroventricular administration, and endovascular administration.

[0102] An effective dose and dosage of pharmaceutical compositions to prevent or treat the disease or condition disclosed herein is defined by an observed beneficial response related to the disease or condition, or symptom of the disease or condition. Beneficial response comprises preventing, alleviating, arresting, or curing the disease or condition, or symptom of the disease or condition. In some embodiments, the beneficial response may be measured by detecting a measurable improvement in the presence, level, or activity, of biomarkers, transcriptomic risk profile, or intestinal microbiome in the subject. An “improvement,” as used herein refers to shift in the presence, level, or activity towards a presence, level, or activity, observed in normal individuals (e.g., individuals who do not suffer from the disease or condition). In instances wherein the therapeutic rAAV composition is not therapeutically effective or is not providing a sufficient alleviation of the disease or condition, or symptom of the disease or condition, then the dosage amount and/or route of administration may be changed, or an additional agent may be administered to the subject, along with the therapeutic rAAV composition. In some embodiments, as a patient is started on a regimen of a therapeutic rAAV composition, the patient is also weaned off (e.g., step-wise decrease in dose) a second treatment regimen.

[0103] In some cases, a dose of the pharmaceutical composition may comprise a concentration of infectious particles of at least or about 10^7 , 10^8 , 10^9 , 10^{10} , 10^{11} , 10^{12} , 10^{13} , 10^{14} , 10^{15} , 10^{16} , or 10^{17} . In some cases, the concentration of infectious particles is 2×10^7 , 2×10^8 , 2×10^9 , 2×10^{10} , 2×10^{11} , 2×10^{12} , 2×10^{13} , 2×10^{14} , 2×10^{15} , 2×10^{16} , or 2×10^{17} . In some cases, the concentration of the infectious particles is 3×10^7 , 3×10^8 , 3×10^9 , 3×10^{10} , 3×10^{11} , 3×10^{12} , 3×10^{13} , 3×10^{14} , 3×10^{15} , 3×10^{16} , or 3×10^{17} . In some cases, the concentration of the infectious particles is 4×10^7 , 4×10^8 , 4×10^9 , 4×10^{10} , 4×10^{11} , 4×10^{12} , 4×10^{13} , 4×10^{14} , 4×10^{15} , 4×10^{16} , or 4×10^{17} . In some cases, the concentration of the infectious particles is 5×10^7 , 5×10^8 , 5×10^9 , 5×10^{10} , 5×10^{11} , 5×10^{12} , 5×10^{13} , 5×10^{14} , 5×10^{15} , 5×10^{16} , or 5×10^{17} . In some cases, the concentration of the infectious particles is 6×10^7 , 6×10^8 , 6×10^9 , 6×10^{10} , 6×10^{11} , 6×10^{12} , 6×10^{13} , 6×10^{14} , 6×10^{15} , 6×10^{16} , or 6×10^{17} . In some cases, the concentration of the infectious particles is 7×10^7 , 7×10^8 , 7×10^9 , 7×10^{10} , 7×10^{11} , 7×10^{12} , 7×10^{13} , 7×10^{14} , 7×10^{15} , 7×10^{16} , or 7×10^{17} . In some cases, the concentration of the infectious particles is 8×10^7 , 8×10^8 , 8×10^9 , 8×10^{10} , 8×10^{11} , 8×10^{12} , 8×10^{13} , 8×10^{14} , 8×10^{15} , 8×10^{16} , or 8×10^{17} . In some cases, the concentration of the infectious particles is 9×10^7 , 9×10^8 , 9×10^9 , 9×10^{10} , 9×10^{11} , 9×10^{12} , 9×10^{13} , 9×10^{14} , 9×10^{15} , 9×10^{16} , or 9×10^{17} .

[0104] Disclosed herein, in some embodiments are formulations of pharmaceutically-acceptable excipients and carrier solutions suitable for delivery of the rAAV compositions described herein, as well as suitable dosing and treatment regimens for using the particular compositions described herein in a variety of treatment regimens. In some embodiments, the amount of therapeutic gene expression product in each therapeutically-useful composition may be prepared in such a way that a suitable dosage will be obtained in any given unit dose of the compound. Factors such as solubility, bioavailability, biological half-life, route of administration, product shelf life, as well as other pharmacological considerations will be contemplated by one skilled in the art of preparing such pharmaceutical formulations, and as such, a variety of dosages and treatment regimens may be desirable.

[0105] In some embodiments, the pharmaceutical forms of the rAAV-based viral compositions suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and/or vegetable oils. Proper fluidity may be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride.

[0106] In some cases, for administration of an injectable aqueous solution, the solution may be suitably buffered, if necessary, and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intramuscular, subcutaneous and intraperitoneal administration. Some variation in dosage will necessarily occur depending on the condition of the subject being treated. The person responsible for administration will, in any event, determine the appropriate dose for the individual subject. Moreover, for human administration, preparations should meet sterility, pyrogenicity, and the general safety and purity standards as required by FDA Office of Biologics standards.

[0107] Disclosed herein are sterile injectable solutions comprising the rAAV compositions disclosed herein, which are prepared by incorporating the rAAV compositions disclosed herein in the required amount in the appropriate solvent with several of the other ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the various sterilized active ingredients into a sterile vehicle which contains the basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum-drying and freeze-drying techniques which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof. Injectable solutions may be advantageous for systemic administration, for example by intravenous or intrathecal administration.

[0108] Suitable dose and dosage administered to a subject is determined by factors including, but not limited to, the particular therapeutic rAAV composition, disease condition and its severity, the identity (e.g., weight, sex, age) of the subject in need of treatment, and can be determined according to the particular circumstances surrounding the case, including, e.g., the specific agent being administered, the route of administration, the condition being treated, and the subject or host being treated.

[0109] The amount of rAAV compositions and time of administration of such compositions will be within the purview of the skilled artisan having benefit of the present teachings. It is likely, however, that the administration of therapeutically-effective amounts of the disclosed compositions may be achieved by a single administration, for example, a single injection of sufficient numbers of infectious particles to provide therapeutic benefit to the patient undergoing such treatment. This is made possible, at least in part, by the fact that certain target cells (e.g., neurons) do not divide, obviating the need for multiple or chronic dosing.

[0110] In certain embodiments, the data obtained from cell culture assays and animal studies are used in formulating the therapeutically effective daily dosage range and/or the therapeutically effective unit dosage amount for use in mammals, including humans. In certain embodiments, the dosage range and/or the unit dosage amount varies within this range depending upon the dosage form employed and the route of administration utilized.

Combination Therapies

[0111] A therapeutic rAAV may be used alone or in combination with an additional therapeutic agent (together, “therapeutic agents”). In some cases, a therapeutic rAAV as used herein is administered alone. The therapeutic agent may be administered together or sequentially in a combination therapy. The combination therapy may be administered within the same day, or may be administered one or more days, weeks, months, or years apart.

[0112] The additional therapeutic agent can comprise a small molecule. The additional therapeutic agent can comprise an antibody, or antigen-binding fragment. The additional therapeutic agent can include lipid nanoparticle-based therapies, anti-sense oligonucleotide therapies, as well as other viral therapies.

[0113] The additional therapeutic agent can comprise a cell-based therapy. Exemplary cell-based therapies include without limitation immune effector cell therapy, chimeric antigen receptor T-cell (CAR-T) therapy, natural killer cell therapy and chimeric antigen receptor natural killer (NK) cell therapy. Either NK cells, or CAR-NK cells, or a combination of both NK cells and CAR-NK cells can be used in combination with the methods disclosed herein. In some embodiments, the NK cells and CAR-NK cells are derived from human induced pluripotent stem cells (iPSC), umbilical cord blood, or a cell line. The NK cells and CAR-NK cells can comprise a cytokine receptor and a suicide gene. The cell-based therapy can comprise a stem cell therapy. The stem cell therapy may be embryonic or somatic stem cells. The stem cells may be isolated from a donor (allogeneic) or isolated from the subject (autologous). The stem cells may be expanded adipose-derived stem cells (eASCs), hematopoietic stem cells (HSCs), mesenchymal stem (stromal) cells (MSCs), or induced pluripotent stem cells (iPSCs) derived from the cells of the subject.

Kits

[0114] Disclosed herein are kits comprising compositions disclosed herein. Also disclosed herein are kits for the treatment or prevention of a disease or conditions of the CNS. In some instances, the disease or condition is cancer, a pathogen infection, pulmonary disease or condition, neurological disease, muscular disease, or an immune disorder, such as those described herein.

[0115] In one embodiment, a kit can include a therapeutic or prophylactic composition containing an effective amount of a composition of a rAAV particle encapsidating a recombinant AAV vector encoding a therapeutic nucleic acid (e.g., therapeutic nucleic acid) and a recombinant AAV (rAAV) capsid protein of the present disclosure. In another embodiment, a kit can include a therapeutic or prophylactic composition containing an effective amount of cells modified by the rAAV described herein (“modified cell”), in unit dosage form that express therapeutic nucleic acid. In some embodiments, a kit comprises a sterile container which can contain a therapeutic composition; such containers can be boxes, ampules, bottles, vials, tubes, bags, pouches, blister-packs, or other suitable container forms known in the art. Such containers can be made of plastic, glass, laminated paper, metal foil, or other materials suitable for holding medications.

[0116] In some instances, the kit further comprises a cell. In some instances, the cell is mammalian. In some instances, the cell is immortalized. In some instances, the immortalized cell is an embryonic stem cell. In some instances, the embryonic stem cell is a human embryonic stem cell. In some instances, the human embryonic stem cell is a human embryonic kidney 293 (HEK-293) cell. In some instances, the kit further comprises an AAV vector comprising a heterologous nucleic acid encoding a therapeutic gene expression product. In some instances, the AAV vector is an episome.

[0117] In some cases, rAAV are provided together with instructions for administering the rAAV to a subject having or at risk of developing the disease or condition (e.g., disease of the CNS). Instructions can generally include information about the use of the composition for the treatment or prevention of the disease or condition.

[0118] In some cases, the instructions include at least one of the following: description of the therapeutic rAAV composition; dosage schedule and administration for treatment or prevention of the disease or condition disclosed herein; precautions; warnings; indications; counter-indications; overdose information; adverse reactions; animal pharmacology; clinical studies; and/or references. The instructions can be printed directly on the container (when present), or as a label applied to the container, or as a separate sheet, pamphlet, card, or folder supplied in or with the container. In some cases, instructions provide procedures for administering the rAAV to the subject alone. In some instances, the instructions provide that the rAAV is formulated for systemic delivery.

Definitions

[0119] The terminology used herein is for the purpose of describing particular cases only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the

extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

[0120] The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the given value. Where particular values are described in the application and claims, unless otherwise stated the term “about” should be assumed to mean an acceptable error range for the particular value.

[0121] As used herein “consisting essentially of” when used to define compositions and methods, shall mean excluding other elements of any essential significance to the combination for the stated purpose. Thus, a composition consisting essentially of the elements as defined herein would not exclude other materials or steps that do not materially affect the basic and novel characteristic(s) of the claimed disclosure, such as compositions for treating skin disorders like acne, eczema, psoriasis, and rosacea.

[0122] The terms “homologous,” “homology,” or “percent homology” are used herein to generally mean an amino acid sequence or a nucleic acid sequence having the same, or similar sequence to a reference sequence. Percent homology of sequences can be determined using the most recent version of BLAST, as of the filing date of this application.

[0123] The terms “increased,” or “increase” are used herein to generally mean an increase by a statically significant amount. In some embodiments, the terms “increased,” or “increase,” mean an increase of at least 10% as compared to a reference level, for example an increase of at least about 10%, at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% increase or any increase between 10-100% as compared to a reference level, standard, or control. Other examples of “increase” include an increase of at least 2-fold, at least 5-fold, at least 10-fold, at least 20-fold, at least 50-fold, at least 100-fold, at least 1000-fold or more as compared to a reference level.

[0124] The terms, “decreased” or “decrease” are used herein generally to mean a decrease by a statistically significant amount. In some embodiments, “decreased” or “decrease” means a reduction by at least 10% as compared to a reference level, for example a decrease by at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% decrease (e.g., absent level or non-detectable level as compared to a reference level), or any decrease between 10-100% as compared to a reference level. In the context of a marker or symptom, by these terms is meant a statistically significant decrease in such level. The decrease can be, for example, at least 10%, at least 20%, at least 30%, at least 40% or more, and is preferably down to a level accepted as within the range of normal for an individual without a given disease.

[0125] The terms “subject” is any organism. In some instances, the organism is a mammal. Non-limiting examples of mammal include, any member of the mamma-

lian class: humans, non-human primates such as chimpanzees, and other apes and monkey species; farm animals such as cattle, horses, sheep, goats, swine; domestic animals such as rabbits, dogs, and cats; laboratory animals including rodents, such as rats, mice and guinea pigs, and the like. In one aspect, the mammal is a human. The term “animal” as used herein comprises human beings and non-human animals. In one embodiment, a “non-human animal” is a mammal, for example a rodent such as rat or a mouse. In one embodiment, a “non-human primate” is a mammal, for example a monkey. In some instances, the subject is a patient, which as used herein, may refer to a subject diagnosed with a particular disease or disorder.

[0126] The term “gene,” as used herein, refers to a segment of nucleic acid that encodes an individual protein or RNA (also referred to as a “coding sequence” or “coding region”), optionally together with associated regulatory region such as promoter, operator, terminator and the like, which may be located upstream or downstream of the coding sequence.

[0127] The term “adeno-associated virus,” or “AAV” as used herein refers to the adeno-associated virus or derivatives thereof. Non-limited examples of AAV’s include AAV type 1 (AAV1), AAV type 2 (AAV2), AAV type 3 (AAV3), AAV type 4 (AAV4), AAV type 5 (AAV5), AAV type 6 (AAV6), AAV type 7 (AAV7), AAV type 8 (AAV8), AAV type 9 (AAV9), AAV type 10 (AAV10), AAV type 11 (AAV11), AAV type 12 (AAV12), avian AAV, bovine AAV, canine AAV, equine AAV, primate AAV, non-primate AAV, and ovine AAV. In some instances, the AAV is described as a “Primate AAV,” which refers to AAV that infect primates. Likewise an AAV may infect bovine animals (e.g., “bovine AAV”, and the like). In some instances, the AAV is wildtype, or naturally occurring. In some instances, the AAV is recombinant.

[0128] The term “AAV capsid” as used herein refers to a capsid protein or peptide of an adeno-associated virus. In some instances, the AAV capsid protein is configured to encapsidate genetic information (e.g., a transgene, therapeutic nucleic acid, viral genome). In some instances, the AAV capsid of the instant disclosure is a modified AAV capsid, relative to a corresponding parental AAV capsid protein.

[0129] The term “tropism” as used herein refers to a quality or characteristic of the AAV capsid that may include an increase or a decrease in enrichment of, expressing the encapsidated genetic information into an in vivo environment, relative to a second in vivo environment. An in vivo environment, in some instances, is a cell-type. An in vivo environment, in some instances, is an organ or organ system.

[0130] The term “AAV vector” as used herein refers to nucleic acid polymer encoding genetic information related to the virus. The AAV vector may be a recombinant AAV vector (rAAV), which refers to an AAV vector generated using recombinatorial genetics methods. In some instances, the rAAV vector comprises at least one heterologous polynucleotide (e.g., a polynucleotide other than a wild-type or naturally occurring AAV genome such as a transgene).

[0131] The term “AAV particle” as used herein refers to an AAV virus, virion, AAV capsid protein or component thereof. In some cases, the AAV particle is modified relative to a parental AAV particle.

[0132] The term “gene product” of “gene expression product” refers to an expression product of a polynucleotide sequence such as, for e.g., a polypeptide, peptide, protein or

RNA, including interfering RNA (e.g., siRNA, miRNA, shRNA) and messenger RNA (mRNA).

[0133] The term “heterologous” as used herein refers to a genetic element (e.g., coding region) or gene expression product (e.g., RNA, protein) that is derived from a genotypically distinct entity from that of the rest of the entity to which it is being compared.

[0134] The term “endogenous” as used herein refers to a genetic element (e.g., coding region) or gene expression product (e.g., RNA, protein) that is naturally occurring in or associated with an organism or a particular cell within the organism.

[0135] The terms “treat,” “treating,” and “treatment” as used herein refers to alleviating or abrogating a disorder, disease, or condition; or one or more of the symptoms associated with the disorder, disease, or condition; or alleviating or eradicating a cause of the disorder, disease, or condition itself. Desirable effects of treatment can include, but are not limited to, preventing occurrence or recurrence of disease, alleviation of symptoms, diminishing any direct or indirect pathological consequences of the disease, preventing metastasis, decreasing the rate of disease progression, amelioration or palliation of the disease state and remission or improved prognosis.

[0136] The term “therapeutically effective amount” refers to the amount of a compound or therapy that, when administered, is sufficient to prevent development of, or alleviate to some extent, one or more of the symptoms of a disorder, disease, or condition of the disease; or the amount of a compound that is sufficient to elicit biological or medical response of a cell, tissue, system, animal, or human that is being sought by a researcher, veterinarian, medical doctor, or clinician.

[0137] The term “pharmaceutically acceptable carrier,” “pharmaceutically acceptable excipient,” “physiologically acceptable carrier,” or “physiologically acceptable excipient” refers to a pharmaceutically acceptable material, composition, or vehicle, such as a liquid or solid filler, diluent, excipient, solvent, or encapsulating material. A component can be “pharmaceutically acceptable” in the sense of being compatible with the other ingredients of a pharmaceutical formulation. It can also be suitable for use in contact with the tissue or organ of humans and animals without excessive toxicity, irritation, allergic response, immunogenicity, or other problems or complications, commensurate with a reasonable benefit/risk ratio. See, Remington: The Science and Practice of Pharmacy, 21st Edition; Lippincott Williams & Wilkins: Philadelphia, PA, 2005; Handbook of Pharmaceutical Excipients, 5th Edition; Rowe et al., Eds., The Pharmaceutical Press and the American Pharmaceutical Association: 2005; and Handbook of Pharmaceutical Additives, 3rd Edition; Ash and Ash Eds., Gower Publishing Company: 2007; Pharmaceutical Preformulation and Formulation, Gibson Ed., CRC Press LLC: Boca Raton, FL, 2004).

[0138] The term “pharmaceutical composition” refers to a mixture of a compound disclosed herein with other chemical components, such as diluents or carriers. The pharmaceutical composition can facilitate administration of the compound to an organism. Multiple techniques of administering a compound exist in the art including, but not limited to, systemic administration.

[0139] Non-limiting examples of “sample” include any material from which nucleic acids and/or proteins can be

obtained. As non-limiting examples, this includes whole blood, peripheral blood, plasma, serum, saliva, mucus, urine, semen, lymph, fecal extract, cheek swab, cells or other bodily fluid or tissue, including but not limited to tissue obtained through surgical biopsy or surgical resection. Alternatively, a sample can be obtained through primary patient derived cell lines, or archived patient samples in the form of preserved samples, or fresh frozen samples.

[0140] The term “in vivo” is used to describe an event that takes place in a subject’s body.

[0141] The term “in vitro” is used to describe an event that takes place contained in a container for holding laboratory reagent such that it is separated from the biological source from which the material is obtained. In vitro assays can encompass cell-based assays in which living or dead cells are employed. In vitro assays can also encompass a cell-free assay in which no intact cells are employed.

[0142] The term “CNS” or “central nervous system” means a tissue selected from brain, thalamus, cortex, putamen, lateral ventricles, medulla, the pons, the amygdala, the motor cortex, caudate, hypothalamus, striatum, ventral mid-brain, neocortex, basal ganglia, hippocampus, cerebellum, cerebellum, brain stem, and spinal cord. The brain includes a variety of cortical and subcortical areas, including the frontal, temporal, occipital and parietal lobes.

[0143] The term “systemic delivery” is defined as a route of administration of medication or other substance into a circulatory system so that the entire body is affected. Administration can take place via enteral administration (absorption of the drug through the gastrointestinal tract) or parenteral administration (generally injection, infusion, or implantation). “Circulatory system” includes both blood or cerebrospinal fluid circulatory systems. Examples of systemic administration for the CNS include intraarterial, intravenous or intrathecal injection. Other examples include administration to the cerebrospinal fluid at any location, in the spine (i.e., but not limited to lumbar) or brain (i.e., but not limited to cisterna magna). The terms “systemic administration” and “systemic delivery” are used interchangeably.

[0144] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

EXAMPLES

Example 1

Method of Identifying the Modified Capsid Proteins in African Green Monkeys

[0145] Of primary concern for the therapeutic applicability of engineered adeno-associated viruses (AAVs) is how well their transduction profiles translate to human application. While previous engineering efforts have focused on in vitro or in vivo rodent screening platforms due to the ease and flexibility of their use, screening efforts directly in non-human primates (NHPs) are much more likely to identify viruses that translate. African Green Monkeys, an old world NHP were selected for the engineering efforts that supported development of the present invention. Engineering efforts were primarily directed on a region of the AAV9 capsid surface located at amino acid position 588, one of the most exposed loops on the capsid surface that is a variable region between natural AAV serotypes and has a role in receptor binding. Insertion of peptides between positions

588 and 589 has been studied in the past by the present inventors, and others, and has resulted in novel receptor binding (AAV-PHP.B/AAV-PHP.eB binding of Ly6a on rodent brain endothelium to facilitate blood-brain barrier crossing and high transduction of the brain) and drastically altered capsid tropism. A library of viral capsids was created by performing a random 7 amino acid insertion at this site within AAV9 in hopes of achieving a novel tropism toward the NHP CNS.

[0146] Plasmids. The first-round viral DNA library was generated by amplification of a section of the AAV9 capsid genome between amino acids 450-599 using NNK degenerate primers (Integrated DNA Technologies, Inc., IDT) to insert seven random amino acids between amino acids 588 and 589 with all possible variations. The resulting library inserts were then introduced into the rAAV-ACap-in-rev-RNA plasmid via Gibson assembly as previously described. The resulting capsid DNA library, rAAV-Cap-Cag-GFP11, contained a diversity of ~1.28 billion variants at the amino acid level. The second round viral DNA library was generated similarly to the first round, but instead of NNK degenerate primers inserted at the 588, a synthesized oligo pool (Twist Bioscience) was used to generate only selected variants in a UBC-Cap-DNA and CAG-Cap-DNA construct with CAP. This second-round DNA library contained a diversity from 3000-15,000 variants at the amino acid level with 2-6 barcoded replicates for each variant.

[0147] The AAV2/9 REP-AAP-ACAP plasmid transfected into HEK293T cells to provide the Rep gene for library viral production prevents production of a wild-type AAV9 capsid during viral library production after a plausible recombination event between this plasmid co-transfected with the library plasmids at each stage containing the library inserts.

[0148] Viral production. Recombinant AAVs were generated according to established protocols. Briefly, immortalized HEK293T cells (ATCC) were quadruple transfected with four vectors using polyethylenimine (PEI). The first vector was the rAAV-Cap-in-cis-Lox library flanked by inverted terminal repeat (ITR) sequences from a parental AAV virus. The second vector was the AAV2/9 REP-AAP-ACAP plasmid. The third vector contains nucleic acids encoding helper virus proteins needed for viral assembly and packaging of the heterologous nucleic acid into the modified capsid structure. The fourth is a pUC-18 plasmid included to achieve the right PEI/DNA ratio for optimal transfection enrichment. Only 10 ng of rAAV-Cap-in-cis-Lox library DNA was transfected (per 150 mm plate) to decrease the likelihood of multiple library DNAs entering the same cell. Viral particles are harvested from the cells and media after 60 h post transfection. Virus present in the media is concentrated by precipitation with 8% polyethylene glycol and 500 mM sodium chloride and the precipitated virus is added to the lysates prepared from the collected cells. The viruses are purified over iodixanol (Optiprep, Sigma) step gradients (15%, 25%, 40%, and 60%). Viruses are concentrated and formulated in PBS. Virus titers are determined by measuring the number of DNaseI-resistant vector genome copies (VGs) using qPCR and the linearized genome plasmid as a control.

[0149] Animals. African Green Monkey procedures were approved by the IACUC committee of Virscio, Inc. African Green monkeys were born and raised in Virscio, Inc. colonies and housed under standard conditions. They were fed ad libitum and received enrichment as part of the primate enrichment program for NHPs at Virscio, Inc. For AAV

infusions, animals were screened for endogenous neutralizing antibodies (nAb). None of the animals that were screened showed any detectable blocking reaction at 1:5 dilution of serum. The day before infusion the animals' food was removed. The test article was administered via intravenous infusion or lumbar intrathecal dosing. Activity and behavior were monitored daily over the course of in-life.

[0150] DNA/RNA recovery and sequencing. Round 1 and round 2 viral libraries were injected into marmosets at a dose of African Green Monkeys at a dose of between $\kappa \times 10^{12}$ to 1×10^{13} vg/kg animal and rAAV genomes were recovered three weeks post injection. Animals were euthanized and brain (both round 1 and round 2), spinal cord (round 1 and round 2) and liver (round 1 and round 2) were recovered, snap frozen, and placed into long-term storage at -80° C. as well as other peripheral tissues such as heart, spleen, adrenal, kidney, and quad. For round 1, the brain was separated into eleven brain regions, 20 mg each, and for round 2, eleven brain regions. 20-300 mg of each brain section, spinal cord, and liver was homogenized in buffer using the Mag-MAX DNA ULTRA (A25597) and a Bead Ruptor 96 (OMNI, INC) and viral DNA was isolated according to the manufacturers recommended protocol. Recovered viral DNA was treated with RNase, and purified with a Zymo DNA Clean and Concentrator kit (D4033). Viral genomes were enriched by 25 cycles of PCR amplification with primers flanking the 588-589 insertion site in the capsid genome using 50% of the total extracted viral DNA as a template. After Zymo DNA purification, samples were diluted 1:10 to 1:1000 depending on tissue type and each dilution further amplified around the library variable region with 10 cycles of PCR. Subsequently, samples were further amplified using custom primers with Illumina Indices for 10 more cycles. The amplification products were run on a 2% low-melting point agarose gel (ThermoFisher Scientific, 16520050) for better separation and recovery of the 210 bp band.

[0151] For the second round library only, packaged viral library DNA was isolated from the injected viral library by digestion of the viral capsid and purification of the contained ssDNA. These viral genomes were amplified by two PCR amplification steps, like the viral DNA extracted from tissue, to add adapters and indices for Illumina next-generation sequencing, and purified after gel electrophoresis. This viral library DNA, along with the viral DNA extracted from tissue, was sent for deep sequencing using an Illumina NextSeq 2000 system.

[0152] NGS data alignment and processing. Raw fastq files from NGS runs were processed with custom-built scripts (Capsida CapSeq Tools). For the first round library, the pipeline to process these datasets involved filtering to remove low-quality reads, utilizing a quality score for each sequence, and eliminating bias from PCR-induced mutations or high GC-content. The filtered dataset was then aligned by a perfect string match algorithm and trimmed to improve the alignment quality. Read counts for each sequence were pulled out and displayed by tissue, at which point all sequences found in the brain were compiled for formation of the second round library.

[0153] For the second round library read counts by tissue were similarly tabulated. Then, a read count of 1 was added to each sequence to remove 0 values, all brain regions for each sequence were summed together, and the read sequences for each codon replicate of a given 7-mer amino

acid sequence were summed together to give a single value for each peptide insertion. Finally, the data was log 10 counts per million (Cpm) normalized. Enrichment values were calculated with normalized cpm of brain vs cpm of virus and converted to log 10.

[0154] While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be

understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

[0155] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

SEQUENCE LISTING

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Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
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Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
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Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
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Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
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Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
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Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 17

Ala Gln Thr Ser Val Val Pro Val Gly Ala Gln
1 5 10

<210> SEQ ID NO 18
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 18

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Ala Gln Tyr Asn Ser Thr Lys Asn Met Ala Gln
1 5 10

<210> SEQ ID NO 19
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 19

Ala Gln Ala Asn Ser Val Lys Asn Ile Ala Gln
1 5 10

<210> SEQ ID NO 20
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 20

Ala Gln Leu Asn Thr Thr Lys Pro Ile Asp Gly
1 5 10

<210> SEQ ID NO 21
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 21

Ala Asn Thr Val Arg Asn Thr
1 5

<210> SEQ ID NO 22
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 22

Asp Asn His Val Arg Thr Val
1 5

<210> SEQ ID NO 23
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 23

Glu Asn Ser Thr Lys Ser Met
1 5

<210> SEQ ID NO 24
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 24

Glu Asn Thr Val Arg Asn Met
1 5

<210> SEQ ID NO 25

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 25

Gly Asn Ala Ile Arg Asn Gly
1 5

<210> SEQ ID NO 26

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 26

Gly Asn Ala Val Lys Pro Phe
1 5

<210> SEQ ID NO 27

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 27

Gly Asn Gly Thr Arg Pro Thr
1 5

<210> SEQ ID NO 28

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 28

His Asn Asn Val Arg Pro Val
1 5

<210> SEQ ID NO 29

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 29

Leu Leu Arg Asp Lys Ser Ser
1 5

<210> SEQ ID NO 30

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 30

Met Asn Asn Thr Ala Val Leu
1 5

<210> SEQ ID NO 31
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 31

Asn Asn Thr Thr Lys Leu Ala
1 5

<210> SEQ ID NO 32
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 32

Asn Asn Thr Val Lys Asn Leu
1 5

<210> SEQ ID NO 33
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 33

Ser Asn Val Ile Arg Asn Ser
1 5

<210> SEQ ID NO 34
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 34

Thr Leu Lys Ser Pro Ala Thr
1 5

<210> SEQ ID NO 35
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 35

Thr Asn Ser Val Lys Leu Ile
1 5

<210> SEQ ID NO 36
<211> LENGTH: 7

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 36

Thr Ser Val Val Pro Val Gly
1 5

<210> SEQ ID NO 37
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 37

Tyr Asn Ser Thr Lys Asn Met
1 5

<210> SEQ ID NO 38
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 38

Ala Asn Ser Val Lys Asn Ile
1 5

<210> SEQ ID NO 39
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 39

Leu Asn Thr Thr Lys Pro Ile
1 5

<210> SEQ ID NO 40
<211> LENGTH: 743
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 40

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20 25 30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85 90 95

-continued

Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	100	105	110
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	115	120	125
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	130	135	140
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	145	150	155
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	165	170	175
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	180	185	190
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	195	200	205
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	210	215	220
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	225	230	235
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	245	250	255
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	260	265	270
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	275	280	285
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	290	295	300
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	305	310	315
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	325	330	335
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	340	345	350
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	355	360	365
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	370	375	380
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	385	390	395
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	405	410	415
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	420	425	430
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	435	440	445
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	450	455	460
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	465	470	475
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	485	490	495

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Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
		515					520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530					535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
				565					570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Asp	Gly	Thr	Leu	Ala	Val
			580					585					590		
Pro	Phe	Lys	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615					620				
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625					630					635					640
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
				645					650					655	
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660					665					670		
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
		675					680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
	690					695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705					710					715					720
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
				725					730					735	
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
			740												

<210> SEQ ID NO 41

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 41

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
			20					25					30		
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
		35					40					45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50					55					60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65					70					75					80
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
			85					90							95

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Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	100	105	110
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	115	120	125
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	130	135	140
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	145	150	155
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	165	170	175
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	180	185	190
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	195	200	205
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	210	215	220
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	225	230	235
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	245	250	255
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	260	265	270
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	275	280	285
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	290	295	300
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	305	310	315
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	325	330	335
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	340	345	350
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	355	360	365
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	370	375	380
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	385	390	395
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	405	410	415
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	420	425	430
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	435	440	445
Arg	Thr	Ile	Lys	Asp	Asn	Thr	Pro	Gly	Arg	Gln	Thr	Leu	Lys	Phe	Ser	450	455	460
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	465	470	475
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	485	490	495
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn			

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500					505					510					
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
515							520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
530					535					540					
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565						570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln
			580					585					590		
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln
		595					600					605			
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His
	610					615					620				
Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Met
	625					630					635				640
Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala
			645						650					655	
Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	Lys	Leu	Asn	Ser	Phe	Ile	Thr
			660					665					670		
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln
		675					680					685			
Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser	Asn
	690					695					700				
Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	Val	Asn	Thr	Glu	Gly	Val
	705					710					715				720
Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn	Leu
			725						730					735	

<210> SEQ ID NO 42
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 42

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1			5						10				15		
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
		20					25					30			
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
	35					40					45				
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50				55					60					
Val	Asn	Ala	Ala	Asp	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	
	65			70				75					80		
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
		85					90						95		
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
		100					105						110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro

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115	120	125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg 130 135 140		
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly 145 150 155 160		
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr 165 170 175		
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro 180 185 190		
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly 195 200 205		
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser 210 215 220		
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile 225 230 235 240		
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu 245 250 255		
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn 260 265 270		
Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg 275 280 285		
Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn 290 295 300		
Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile 305 310 315 320		
Gln Val Lys Glu Val Thr Asp Asn Asn Gly Val Lys Thr Ile Ala Asn 325 330 335		
Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu 340 345 350		
Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro 355 360 365		
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp 370 375 380		
Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe 385 390 395 400		
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu 405 410 415		
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu 420 425 430		
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser 435 440 445		
Arg Thr Ile Gln Gln Gly Lys Gln Ser Val Gln Thr Leu Lys Phe Ser 450 455 460		
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro 465 470 475 480		
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn 485 490 495		
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn 500 505 510		
Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys 515 520 525		

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Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly
 530 535 540
 Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile
 545 550 555 560
 Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser
 565 570 575
 Tyr Gly Gln Val Ala Thr Asn His Gln Ser Asp Gly Thr Leu Ala Val
 580 585 590
 Pro Phe Lys Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
 595 600 605
 Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 43
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 43

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125

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Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
130						135					140				
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly
145					150					155					160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr
				165					170					175	
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180					185					190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
		195					200					205			
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
	210					215					220				
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225					230					235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
		260						265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
	290					295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
			325						330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
		340						345					350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355					360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
	370					375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390					395					400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
			405					410						415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
		420						425					430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
		435				440					445				
Arg	Thr	Ile	Asp	Gly	Ala	Ala	Thr	Lys	Asn	Gln	Thr	Leu	Lys	Phe	Ser
	450					455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
				485					490					495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
	515						520					525			

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Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530					535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
				565					570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Asp	Gly	Thr	Leu	Ala	Val
			580					585					590		
Pro	Phe	Lys	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615					620				
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625					630					635					640
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
				645					650					655	
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660					665					670		
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
	675						680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
	690					695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705					710					715					720
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
				725					730					735	
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
			740												

<210> SEQ ID NO 44
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 44

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
		20						25					30		
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
		35				40						45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50				55						60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65				70					75					80	
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
			85					90						95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
		100						105					110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
	115						120					125			

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Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
130						135					140				
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly
145					150					155					160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr
				165					170					175	
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180					185					190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
		195						200				205			
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
		210				215					220				
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225					230					235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
		260						265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
		290				295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
				325					330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340					345					350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355					360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
		370				375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390					395					400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
			405					410						415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
			420				425						430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
		435				440					445				
Arg	Thr	Ile	Asp	Gly	Gln	Ser	Ser	Lys	Ser	Gln	Thr	Leu	Lys	Phe	Ser
		450				455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
				485				490						495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500				505						510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
		515					520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly

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530					535					540					
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565						570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Asp	Gly	Thr	Leu	Ala	Val
			580					585					590		
Pro	Phe	Lys	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615				620					
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625						630				635				640	
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
			645						650				655		
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660				665					670			
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
		675					680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
	690					695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705						710				715				720	
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
			725						730				735		
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
			740												

<210> SEQ ID NO 45
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 45

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
			20					25					30		
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
		35					40					45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50					55					60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65					70					75					80
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
			85						90					95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
			100					105					110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
		115					120					125			
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg

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130					135					140					
Pro 145	Val	Glu	Gln	Ser	Pro 150	Gln	Glu	Pro	Asp	Ser 155	Ser	Ala	Gly	Ile	Gly 160
Lys	Ser	Gly	Ala	Gln 165	Pro	Ala	Lys	Lys	Arg 170	Leu	Asn	Phe	Gly	Gln 175	Thr
Gly	Asp	Thr	Glu 180	Ser	Val	Pro	Asp	Pro 185	Gln	Pro	Ile	Gly	Glu 190	Pro	Pro
Ala	Ala	Pro 195	Ser	Gly	Val	Gly	Ser 200	Leu	Thr	Met	Ala	Ser 205	Gly	Gly	Gly
Ala	Pro 210	Val	Ala	Asp	Asn 215	Asn	Glu	Gly	Ala	Asp	Gly 220	Val	Gly	Ser	Ser
Ser 225	Gly	Asn	Trp	His 230	Cys	Asp	Ser	Gln	Trp	Leu 235	Gly	Asp	Arg	Val	Ile 240
Thr	Thr	Ser	Thr 245	Arg	Thr	Trp	Ala	Leu	Pro 250	Thr	Tyr	Asn	Asn	His 255	Leu
Tyr	Lys	Gln 260	Ile	Ser	Asn	Ser	Thr 265	Ser	Gly	Gly	Ser	Ser	Asn 270	Asp	Asn
Ala	Tyr	Phe 275	Gly	Tyr	Ser	Thr	Pro 280	Trp	Gly	Tyr	Phe	Asp 285	Phe	Asn	Arg
Phe 290	His	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln	Arg 300	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn	Phe 315	Lys	Leu	Phe	Asn	Ile 320
Gln	Val	Lys	Glu 325	Val	Thr	Asp	Asn	Asn	Gly 330	Val	Lys	Thr	Ile	Ala 335	Asn
Asn	Leu	Thr 340	Ser	Thr	Val	Gln	Val	Phe 345	Thr	Asp	Ser	Asp	Tyr 350	Gln	Leu
Pro	Tyr 355	Val	Leu	Gly	Ser	Ala	His 360	Glu	Gly	Cys	Leu	Pro 365	Pro	Phe	Pro
Ala	Asp 370	Val	Phe	Met	Ile	Pro 375	Gln	Tyr	Gly	Tyr	Leu	Thr 380	Leu	Asn	Asp
Gly 385	Ser	Gln	Ala	Val	Gly 390	Arg	Ser	Ser	Phe	Tyr 395	Cys	Leu	Glu	Tyr	Phe 400
Pro	Ser	Gln	Met 405	Leu	Arg	Thr	Gly	Asn	Asn 410	Phe	Gln	Phe	Ser	Tyr 415	Glu
Phe	Glu 420	Asn	Val	Pro	Phe	His	Ser	Ser 425	Tyr	Ala	His	Ser	Gln 430	Ser	Leu
Asp	Arg 435	Leu	Met	Asn	Pro	Leu	Ile 440	Asp	Gln	Tyr	Leu	Tyr 445	Tyr	Leu	Ser
Arg 450	Thr	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln	Gln	Thr 460	Leu	Lys	Phe	Ser
Val 465	Ala	Gly	Pro	Ser	Asn 470	Met	Ala	Val	Gln	Gly 475	Arg	Asn	Tyr	Ile	Pro 480
Gly	Pro	Ser	Tyr 485	Arg	Gln	Gln	Arg	Val	Ser 490	Thr	Thr	Val	Thr	Gln 495	Asn
Asn	Asn	Ser 500	Glu	Phe	Ala	Trp	Pro	Gly 505	Ala	Ser	Ser	Trp	Ala 510	Leu	Asn
Gly	Arg 515	Asn	Ser	Leu	Met	Asn	Pro 520	Gly	Pro	Ala	Met 525	Ala	Ser	His	Lys
Glu 530	Gly	Glu	Asp	Arg	Phe	Phe 535	Pro	Leu	Ser	Gly 540	Ser	Leu	Ile	Phe	Gly

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Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile
 545 550 555 560
 Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser
 565 570 575
 Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Asn Thr Val
 580 585 590
 Arg Asn Thr Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
 595 600 605
 Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 46
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 46

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140

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Pro Val Glu Gln Ser	Pro Gln Glu Pro Asp	Ser Ser Ala Gly Ile Gly
145	150	155 160
Lys Ser Gly Ala Gln Pro	Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	
165	170	175
Gly Asp Thr Glu Ser Val Pro	Asp Pro Gln Pro Ile Gly Glu Pro Pro	
180	185	190
Ala Ala Pro Ser Gly Val Gly	Ser Leu Thr Met Ala Ser Gly Gly Gly	
195	200	205
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp	Gly Val Gly Ser Ser	
210	215	220
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile		
225	230	235 240
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu		
245	250	255
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn		
260	265	270
Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg		
275	280	285
Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn		
290	295	300
Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile		
305	310	315 320
Gln Val Lys Glu Val Thr Asp Asn Asn Gly Val Lys Thr Ile Ala Asn		
325	330	335
Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu		
340	345	350
Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro		
355	360	365
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp		
370	375	380
Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe		
385	390	395 400
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu		
405	410	415
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu		
420	425	430
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser		
435	440	445
Arg Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser		
450	455	460
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro		
465	470	475 480
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn		
485	490	495
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn		
500	505	510
Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys		
515	520	525
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly		
530	535	540

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Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
				565					570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Asp	Asn	His	Val
		580						585					590		
Arg	Thr	Val	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615					620				
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625					630					635					640
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
			645						650					655	
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660					665					670		
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
		675					680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
	690					695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705					710					715					720
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
			725						730					735	
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
			740												

<210> SEQ ID NO 47
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 47

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
		20						25					30		
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
		35				40						45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50				55						60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65				70					75					80	
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
			85					90						95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
		100						105					110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
	115						120					125			
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
	130					135					140				

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Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly		
145					150					155					160		
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr		
				165					170					175			
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro		
			180					185					190				
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly		
		195						200				205					
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser		
	210					215					220						
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile		
225					230					235					240		
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu		
				245					250					255			
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn		
		260						265					270				
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg		
		275					280					285					
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn		
	290					295					300						
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile		
305					310					315					320		
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn		
				325					330					335			
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu		
			340					345					350				
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro		
		355					360					365					
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp		
	370					375					380						
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe		
385					390					395					400		
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu		
			405					410						415			
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu		
			420				425						430				
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser		
		435					440					445					
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser		
	450					455					460						
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro		
465					470					475					480		
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn		
				485					490					495			
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn		
			500					505					510				
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys		
		515					520					525					
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly		
	530					535					540						
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile		

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<210> SEQ ID NO 48
<211> LENGTH: 743
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 48

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
          20          25          30
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
          35          40          45
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
          50          55          60
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65          70          75          80
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
          85          90          95
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
          100          105          110
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
          115          120          125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
          130          135          140
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly

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145					150						155					160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	
				165					170					175		
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	
			180					185					190			
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	
		195					200					205				
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	
	210					215					220					
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
225					230					235					240	
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
				245					250						255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260						265					270			
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
	290					295					300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
305					310					315					320	
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
		340						345					350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355					360					365				
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
	370					375					380					
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
385					390					395					400	
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	
			405					410							415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
		420						425					430			
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	
	435						440					445				
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	
	450					455					460					
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	
465					470					475					480	
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
			485					490						495		
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	
			500					505					510			
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
		515					520					525				
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	
	530					535					540					
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	
545					550					555					560	

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Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser
 565 570 575
 Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Glu Asn Thr Val
 580 585 590
 Arg Asn Met Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
 595 600 605
 Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 49
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 49

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160

Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr
				165					170					175	
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180					185					190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
		195					200				205				
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
		210				215					220				
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225					230					235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
			260					265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280				285				
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
		290				295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
			325						330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340					345					350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355					360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
		370				375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390				395						400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
				405					410					415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
			420					425					430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
		435					440					445			
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
		450				455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
					470				475					480	
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
				485				490						495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	S												

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Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	
				565					570					575		
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Leu	Asn	Thr	Thr	
			580					585					590			
Lys	Pro	Ile	Asp	Gly	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	
		595					600					605				
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	
	610					615					620					
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	
625					630					635					640	
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	
				645					650					655		
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	
			660					665					670			
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	
	675						680					685				
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	
	690					695					700					
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	
705					710					715					720	
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	
				725					730					735		
Arg	Tyr	Leu	Thr	Arg	Asn	Leu										
			740													

<210> SEQ ID NO 50
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 50

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser	
1				5				10					15			
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro	
		20					25					30				
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro	
	35					40					45					
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro	
	50				55					60						
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	
65				70					75					80		
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	
		85						90					95			
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
		100					105						110			
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	
	115						120					125				
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	
	130				135						140					
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	
145				150					155					160		

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Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr		
				165					170					175			
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro		
			180					185					190				
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly		
		195					200					205					
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser		
		210				215					220						
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile		
225				230						235					240		
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu		
			245						250					255			
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn		
		260						265					270				
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg		
		275					280					285					
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn		
	290					295					300						
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile		
305					310					315					320		
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn		
			325						330					335			
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu		
		340						345					350				
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro		
		355					360					365					
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp		
		370				375					380						
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe		
385					390					395					400		
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu		
			405					410						415			
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu		
		420					425						430				
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser		
		435				440						445					
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser		
		450				455					460						
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro		
465					470					475					480		
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn		
			485					490						495			
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn		
		500						505					510				
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys		
		515					520					525					
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly		
		530				535					540						
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile		
545					550					555					560		
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser		

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565					570					575					
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Gly	Asn	Ala	Ile
			580					585					590		
Arg	Asn	Gly	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615					620				
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625						630					635				640
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
			645						650					655	
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660				665						670		
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
		675					680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
	690					695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705						710					715				720
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
			725						730					735	
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
			740												

<210> SEQ ID NO 51
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 51

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
		20					25						30		
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
		35					40					45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50					55					60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65				70						75					80
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
		85						90						95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
		100						105					110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
		115					120					125			
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
	130					135					140				
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly
145					150					155					160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr

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165							170							175						
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro					
			180							185				190						
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly					
			195				200						205							
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser					
			210				215						220							
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile					
225				230						235			240							
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu					
			245						250			255								
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn					
			260						265			270								
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg					
			275			280						285								
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn					
			290			295						300								
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile					
305				310						315			320							
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn					
			325						330			335								
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu					
			340			345						350								
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro					
			355			360						365								
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp					
			370			375						380								
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe					
385				390						395			400							
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu					
			405						410			415								
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu					
			420			425						430								
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser					
			435			440						445								
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser					
			450			455						460								
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro					
465				470						475			480							
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn					
			485						490			495								
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn					
			500			505						510								
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys					
			515			520						525								
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly					
			530			535						540								
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile					
545				550						555			560							
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser					
			565			570						575								

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Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ser Asn Val Ile
 580 585 590
 Arg Asn Ser Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
 595 600 605
 Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 52
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 52

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175

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Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	180	185	190
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	195	200	205
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	210	215	220
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	225	230	235
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	245	250	255
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	260	265	270
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	275	280	285
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	290	295	300
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	305	310	315
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	325	330	335
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	340	345	350
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	355	360	365
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	370	375	380
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	385	390	395
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	405	410	415
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	420	425	430
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	435	440	445
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	450	455	460
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	465	470	475
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	485	490	495
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	500	505	510
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	515	520	525
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	530	535	540
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	545	550	555
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	565	570	575

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Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Asn Ser Val
 580 585 590
 Lys Asn Ile Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
 595 600 605
 Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 53
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 53

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175

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Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180					185					190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
		195					200					205			
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
	210				215						220				
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225				230						235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
			245						250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
		260						265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
	275						280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
	290					295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
			325						330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340					345					350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355					360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
	370					375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390					395					400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
			405						410					415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
			420					425					430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
		435					440					445			
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
	450					455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
			485						490					495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
	515						520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530					535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565					570						575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Tyr	Asn	Ser	Thr

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580	585	590
Lys Asn Met Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile		
595	600	605
Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro		
610	615	620
Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro		
625	630	635
Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile		
645	650	655
Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp		
660	665	670
Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val		
675	680	685
Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro		
690	695	700
Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe		
705	710	715
Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr		
725	730	735
Arg Tyr Leu Thr Arg Asn Leu		
740		

<210> SEQ ID NO 54

<211> LENGTH: 743

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 54

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser		
1	5	10
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro		
20	25	30
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro		
35	40	45
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro		
50	55	60
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp		
65	70	75
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala		
85	90	95
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly		
100	105	110
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro		
115	120	125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg		
130	135	140
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly		
145	150	155
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr		
165	170	175
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro		

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180					185					190				
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly
	195						200					205		
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser
	210					215					220			Ser
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val
225					230					235				Ile
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His
				245					250					255
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp
		260						265					270	Asn
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn
		275					280					285		Arg
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn
290						295					300			Asn
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn
305					310					315				Ile
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala
			325						330					335
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln
		340						345					350	Leu
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe
		355					360					365		Pro
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn
	370					375					380			Asp
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr
385					390					395				Phe
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr
			405					410						Glu
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser
		420					425						430	Leu
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu
		435				440						445		Ser
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe
	450					455					460			Ser
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile
465					470					475				Pro
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln
				485				490						495
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu
		500						505					510	Asn
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His
		515					520					525		Lys
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe
	530					535					540			Gly
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met
545					550					555				Ile
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu
				565				570						575
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Thr	Leu	Lys
			580					585						Ser

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Pro Ala Thr Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
   595                               600                               605

Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
   610                               615                               620

Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
   625                               630                               635                               640

Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
                               645                               650                               655

Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
   660                               665                               670

Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
   675                               680                               685

Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
   690                               695                               700

Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
   705                               710                               715                               720

Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
   725                               730                               735

Arg Tyr Leu Thr Arg Asn Leu
   740

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<210> SEQ ID NO 55
<211> LENGTH: 743
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 55

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Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1           5           10           15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20           25           30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35           40           45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50           55           60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65           70           75           80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85           90           95

Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100          105          110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115          120          125

Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130          135          140

Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145          150          155          160

Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
165          170          175

Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
180          185          190

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Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
195		200													
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
210		215													
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225		230													
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
		245													
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
		260													
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275													
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
290		295													
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305		310													
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
		325													
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
		340													
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
355		360													
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
370		375													
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385		390													
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
		405													
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
		420													
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
435		440													
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
450		455													
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465		470													
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
		485													
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
		500													
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
515		520													
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
530		535													
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545		550													
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
		565													
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Gly	Asn	Ala	Val
		580													

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Lys Pro Phe Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile
 595 600 605
 Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 56
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 56

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175
 Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
 180 185 190

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Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	195	200	205	
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	210	215	220	
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	225	230	235	240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	245	250	255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	260	265	270	
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	275	280	285	
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	290	295	300	
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	305	310	315	320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	325	330	335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	340	345	350	
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	355	360	365	
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	370	375	380	
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	385	390	395	400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	405	410	415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	420	425	430	
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	435	440	445	
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	450	455	460	
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	465	470	475	480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	485	490	495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	500	505	510	
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	515	520	525	
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	530	535	540	
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	545	550	555	560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	565	570	575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Gly	Asn	Gly	Thr	580	585	590	
Arg	Pro	Thr	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile				

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595	600	605
Leu Pro Gly Met Val Trp	Gln Asp Arg Asp Val Tyr	Leu Gln Gly Pro
610	615	620
Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro		
625	630	635 640
Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile		
	645	650 655
Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp		
	660	665 670
Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val		
	675	680 685
Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro		
	690	695 700
Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe		
	705	710 715 720
Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr		
	725	730 735
Arg Tyr Leu Thr Arg Asn Leu		
	740	

<210> SEQ ID NO 57
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 57

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20 25 30
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35 40 45
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85 90 95
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100 105 110
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115 120 125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130 135 140
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145 150 155 160
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
165 170 175
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
180 185 190
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly

-continued

195					200					205					
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
210					215						220				
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
225					230					235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
			260						265				270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
290						295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
				325					330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340						345				350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355					360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
370						375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390					395					400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
				405					410					415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
			420						425				430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
		435					440					445			
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
		450				455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
				485					490					495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
		515					520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
		530				535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
				565					570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	His	Asn	Asn	Val
			580					585					590		
Arg	Pro	Val	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595					600					605			

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Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro
 610 615 620
 Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
 625 630 635 640
 Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 58
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 58

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175
 Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
 180 185 190
 Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
 195 200 205

Ala 210	Pro	Val	Ala	Asp	Asn	Asn 215	Glu	Gly	Ala	Asp	Gly 220	Val	Gly	Ser	Ser
Ser 225	Gly	Asn	Trp	His	Cys 230	Asp	Ser	Gln	Trp	Leu 235	Gly	Asp	Arg	Val	Ile 240
Thr	Thr	Ser	Thr	Arg 245	Thr	Trp	Ala	Leu	Pro 250	Thr	Tyr	Asn	Asn	His 255	Leu
Tyr	Lys	Gln	Ile 260	Ser	Asn	Ser	Thr	Ser	Gly 265	Gly	Ser	Ser	Asn	Asp	Asn
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro 280	Trp	Gly	Tyr	Phe	Asp 285	Phe	Asn	Arg
Phe 290	His	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln	Arg 300	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn	Phe 315	Lys	Leu	Phe	Asn	Ile 320
Gln	Val	Lys	Glu 325	Val	Thr	Asp	Asn	Asn	Gly 330	Val	Lys	Thr	Ile	Ala 335	Asn
Asn	Leu	Thr	Ser 340	Thr	Val	Gln	Val	Phe	Thr 345	Asp	Ser	Asp	Tyr	Gln	Leu
Pro	Tyr	Val 355	Leu	Gly	Ser	Ala 360	His	Glu	Gly	Cys	Leu 365	Pro	Pro	Phe	Pro
Ala 370	Asp	Val	Phe	Met	Ile	Pro 375	Gln	Tyr	Gly	Tyr	Leu 380	Thr	Leu	Asn	Asp
Gly 385	Ser	Gln	Ala	Val	Gly 390	Arg	Ser	Ser	Phe	Tyr 395	Cys	Leu	Glu	Tyr	Phe 400
Pro	Ser	Gln	Met 405	Leu	Arg	Thr	Gly	Asn	Asn 410	Phe	Gln	Phe	Ser	Tyr 415	Glu
Phe	Glu	Asn	Val 420	Pro	Phe	His	Ser	Ser	Tyr 425	Ala	His	Ser	Gln	Ser	Leu
Asp	Arg	Leu 435	Met	Asn	Pro	Leu 440	Ile	Asp	Gln 445	Tyr	Leu	Tyr 445	Tyr	Leu	Ser
Arg 450	Thr	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln 460	Gln	Thr	Leu	Lys	Phe	Ser
Val 465	Ala	Gly	Pro	Ser	Asn 470	Met	Ala	Val	Gln 475	Gly	Arg	Asn	Tyr	Ile	Pro 480
Gly	Pro	Ser	Tyr 485	Arg	Gln	Gln	Arg	Val	Ser 490	Thr	Thr	Val	Thr	Gln 495	Asn
Asn	Asn	Ser	Glu 500	Phe	Ala	Trp	Pro	Gly 505	Ala	Ser	Ser	Trp	Ala	Leu	Asn
Gly	Arg	Asn 515	Ser	Leu	Met	Asn 520	Pro	Gly	Pro	Ala	Met	Ala 525	Ser	His	Lys
Glu 530	Gly	Glu	Asp	Arg	Phe	Phe 535	Pro	Leu	Ser	Gly	Ser	Leu 540	Ile	Phe	Gly
Lys 545	Gln	Gly	Thr	Gly	Arg 550	Asp	Asn	Val	Asp	Ala 555	Asp	Lys	Val	Met	Ile 560
Thr	Asn	Glu	Glu 565	Glu	Ile	Lys	Thr	Thr	Asn 570	Pro	Val	Ala	Thr	Glu 575	Ser
Tyr	Gly	Gln 580	Val	Ala	Thr	Asn	His	Gln 585	Ser	Ala	Gln	Leu 590	Leu	Arg	Asp
Lys	Ser	Ser 595	Ala	Gln	Ala	Gln 600	Thr	Gly	Trp	Val	Gln	Asn 605	Gln	Gly	Ile

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Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
610						615					620				
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625					630					635					640
Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile
				645					650					655	
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp
			660					665					670		
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val
		675					680					685			
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro
		690				695					700				
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe
705					710					715					720
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr
				725					730					735	
Arg	Tyr	Leu	Thr	Arg	Asn	Leu									
				740											

<210> SEQ ID NO 59
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 59

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
		20						25					30		
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
		35					40					45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
		50				55					60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65					70					75					80
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
			85						90					95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
			100					105					110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
		115					120					125			
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
		130				135					140				
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly
145					150					155					160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr
			165						170					175	
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180					185					190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
		195					200						205		

Ala 210	Pro	Val	Ala	Asp	Asn	Asn 215	Glu	Gly	Ala	Asp	Gly 220	Val	Gly	Ser	Ser
Ser 225	Gly	Asn	Trp	His	Cys 230	Asp	Ser	Gln	Trp	Leu 235	Gly	Asp	Arg	Val	Ile 240
Thr	Thr	Ser	Thr	Arg 245	Thr	Trp	Ala	Leu	Pro 250	Thr	Tyr	Asn	Asn	His 255	Leu
Tyr	Lys	Gln	Ile 260	Ser	Asn	Ser	Thr	Ser 265	Gly	Gly	Ser	Ser	Asn	Asp	Asn
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro 280	Trp	Gly	Tyr	Phe	Asp 285	Phe	Asn	Arg
Phe 290	His	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln	Arg 300	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn	Phe 315	Lys	Leu	Phe	Asn	Ile 320
Gln	Val	Lys	Glu 325	Val	Thr	Asp	Asn	Asn	Gly 330	Val	Lys	Thr	Ile	Ala 335	Asn
Asn	Leu	Thr	Ser 340	Thr	Val	Gln	Val	Phe 345	Thr	Asp	Ser	Asp	Tyr 350	Gln	Leu
Pro	Tyr	Val 355	Leu	Gly	Ser	Ala	His 360	Glu	Gly	Cys	Leu	Pro 365	Pro	Phe	Pro
Ala 370	Asp	Val	Phe	Met	Ile	Pro 375	Gln	Tyr	Gly	Tyr	Leu 380	Thr	Leu	Asn	Asp
Gly 385	Ser	Gln	Ala	Val	Gly 390	Arg	Ser	Ser	Phe	Tyr 395	Cys	Leu	Glu	Tyr	Phe 400
Pro	Ser	Gln	Met 405	Leu	Arg	Thr	Gly	Asn	Asn 410	Phe	Gln	Phe	Ser	Tyr 415	Glu
Phe	Glu	Asn 420	Val	Pro	Phe	His	Ser	Ser 425	Tyr	Ala	His	Ser	Gln	Ser	Leu
Asp	Arg	Leu 435	Met	Asn	Pro	Leu	Ile 440	Asp	Gln	Tyr	Leu 445	Tyr	Tyr	Leu	Ser
Arg 450	Thr	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln	Gln	Thr 460	Leu	Lys	Phe	Ser
Val 465	Ala	Gly	Pro	Ser	Asn 470	Met	Ala	Val	Gln	Gly 475	Arg	Asn	Tyr	Ile	Pro 480
Gly	Pro	Ser	Tyr 485	Arg	Gln	Gln	Arg	Val	Ser 490	Thr	Thr	Val	Thr	Gln 495	Asn
Asn	Asn	Ser 500	Glu	Phe	Ala	Trp	Pro	Gly 505	Ala	Ser	Ser	Trp	Ala 510	Leu	Asn
Gly	Arg	Asn 515	Ser	Leu	Met	Asn	Pro 520	Gly	Pro	Ala	Met 525	Ala	Ser	His	Lys
Glu 530	Gly	Glu	Asp	Arg	Phe	Phe 535	Pro	Leu	Ser	Gly	Ser 540	Leu	Ile	Phe	Gly
Lys 545	Gln	Gly	Thr	Gly	Arg 550	Asp	Asn	Val	Asp	Ala 555	Asp	Lys	Val	Met	Ile 560
Thr	Asn	Glu	Glu 565	Glu	Ile	Lys	Thr	Thr	Asn 570	Pro	Val	Ala	Thr	Glu 575	Ser
Tyr	Gly	Gln 580	Val	Ala	Thr	Asn	His	Gln 585	Ser	Ala	Gln	Met 590	Asn	Asn	Thr
Ala 595	Val	Leu	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln 605	Asn	Gln	Gly	Ile
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro

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610	615	620
Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro		
625	630	635 640
Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile		
	645	650 655
Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp		
	660	665 670
Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val		
	675	680 685
Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro		
	690	695 700
Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe		
	705	710 715 720
Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr		
	725	730 735
Arg Tyr Leu Thr Arg Asn Leu		
	740	
<210> SEQ ID NO 60		
<211> LENGTH: 743		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Synthesized		
<400> SEQUENCE: 60		
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser		
1	5	10 15
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro		
	20	25 30
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro		
	35	40 45
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro		
	50	55 60
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp		
	65	70 75 80
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala		
	85	90 95
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly		
	100	105 110
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro		
	115	120 125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg		
	130	135 140
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly		
	145	150 155 160
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr		
	165	170 175
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro		
	180	185 190
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly		
	195	200 205
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser		

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210					215					220					
Ser 225	Gly	Asn	Trp	His	Cys 230	Asp	Ser	Gln	Trp	Leu 235	Gly	Asp	Arg	Val	Ile 240
Thr	Thr	Ser	Thr	Arg 245	Thr	Trp	Ala	Leu	Pro 250	Thr	Tyr	Asn	Asn	His 255	Leu
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly 265	Gly	Ser	Ser	Asn	Asp	Asn
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro 280	Trp	Gly	Tyr	Phe	Asp 285	Phe	Asn	Arg
Phe	His 290	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln	Arg 300	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn	Phe 315	Lys	Leu	Phe	Asn	Ile 320
Gln	Val	Lys	Glu	Val 325	Thr	Asp	Asn	Asn	Gly 330	Val	Lys	Thr	Ile	Ala 335	Asn
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe 345	Thr	Asp	Ser	Asp	Tyr 350	Gln	Leu
Pro	Tyr	Val 355	Leu	Gly	Ser	Ala	His 360	Glu	Gly	Cys	Leu	Pro 365	Pro	Phe	Pro
Ala	Asp 370	Val	Phe	Met	Ile	Pro 375	Gln	Tyr	Gly	Tyr	Leu 380	Thr	Leu	Asn	Asp
Gly 385	Ser	Gln	Ala	Val	Gly 390	Arg	Ser	Ser	Phe	Tyr 395	Cys	Leu	Glu	Tyr	Phe 400
Pro	Ser	Gln	Met	Leu 405	Arg	Thr	Gly	Asn	Asn	Phe 410	Gln	Phe	Ser	Tyr 415	Glu
Phe	Glu	Asn	Val 420	Pro	Phe	His	Ser	Ser	Tyr 425	Ala	His	Ser	Gln	Ser	Leu
Asp	Arg	Leu 435	Met	Asn	Pro	Leu	Ile 440	Asp	Gln	Tyr	Leu 445	Tyr	Tyr	Leu	Ser
Arg	Thr 450	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln	Gln	Thr 460	Leu	Lys	Phe	Ser
Val 465	Ala	Gly	Pro	Ser	Asn 470	Met	Ala	Val	Gln	Gly 475	Arg	Asn	Tyr	Ile	Pro 480
Gly	Pro	Ser	Tyr	Arg 485	Gln	Gln	Arg	Val	Ser 490	Thr	Thr	Val	Thr	Gln 495	Asn
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly 505	Ala	Ser	Ser	Trp	Ala	Leu	Asn
Gly	Arg	Asn 515	Ser	Leu	Met	Asn	Pro 520	Gly	Pro	Ala	Met	Ala 525	Ser	His	Lys
Glu	Gly 530	Glu	Asp	Arg	Phe	Phe 535	Pro	Leu	Ser	Gly	Ser 540	Leu	Ile	Phe	Gly
Lys 545	Gln	Gly	Thr	Gly	Arg 550	Asp	Asn	Val	Asp	Ala 555	Asp	Lys	Val	Met	Ile 560
Thr	Asn	Glu	Glu	Glu 565	Ile	Lys	Thr	Thr	Asn 570	Pro	Val	Ala	Thr	Glu 575	Ser
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln 585	Ser	Ala	Gln	Asn	Asn	Thr	Thr
Lys	Leu	Ala 595	Ala	Gln	Ala	Gln	Thr 600	Gly	Trp	Val	Gln 605	Asn	Gln	Gly	Ile
Leu 610	Pro	Gly	Met	Val	Trp	Gln 615	Asp	Arg	Asp	Val	Tyr 620	Leu	Gln	Gly	Pro

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Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
625                630                635                640

Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
                645                650                655

Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
                660                665                670

Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
                675                680                685

Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
        690                695                700

Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
705                710                715                720

Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
                725                730                735

Arg Tyr Leu Thr Arg Asn Leu
        740

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<210> SEQ ID NO 61
<211> LENGTH: 743
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 61

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Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20         25         30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35         40         45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50         55         60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65         70         75         80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85         90         95

Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100        105        110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115        120        125

Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130        135        140

Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145        150        155        160

Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
165        170        175

Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
180        185        190

Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
195        200        205

Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
210        215        220

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Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
225					230					235					240	
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
				245					250					255		
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260						265					270			
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
	275						280					285				
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
290						295					300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
305					310					315					320	
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
		340						345					350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
	355					360					365					
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
	370					375					380					
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
385					390					395					400	
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	
			405					410						415		
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
		420						425					430			
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	
		435					440					445				
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	
	450					455					460					
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	
465					470					475					480	
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
			485					490						495		
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	
		500						505					510			
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
	515					520						525				
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	
	530				535						540					
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	
545					550					555					560	
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	
			565					570						575		
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Asn	Asn	Thr	Val	
		580						585					590			
Lys	Asn	Leu	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	
	595					600						605				
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	
610						615					620					

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Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro
625                      630                      635                      640

Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
                      645                      650                      655

Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
                      660                      665                      670

Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
                      675                      680                      685

Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
690                      695                      700

Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
705                      710                      715                      720

Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
                      725                      730                      735

Arg Tyr Leu Thr Arg Asn Leu
740

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<210> SEQ ID NO 62
<211> LENGTH: 743
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 62

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Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20          25          30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35          40          45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50          55          60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65          70          75          80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85          90          95

Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100         105         110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115         120         125

Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130         135         140

Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145         150         155         160

Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
165         170         175

Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
180         185         190

Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
195         200         205

Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
210         215         220

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Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	225	230	235	240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	245	250	255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	260	265	270	
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	275	280	285	
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	290	295	300	
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	305	310	315	320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	325	330	335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	340	345	350	
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	355	360	365	
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	370	375	380	
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	385	390	395	400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	405	410	415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	420	425	430	
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	435	440	445	
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	450	455	460	
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	465	470	475	480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	485	490	495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	500	505	510	
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	515	520	525	
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	530	535	540	
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	545	550	555	560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	565	570	575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ser	Asn	Val	Ile	580	585	590	
Arg	Asn	Ser	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	595	600	605	
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	610	615	620	
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro				

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625	630	635	640
Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile	645	650	655
Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp	660	665	670
Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val	675	680	685
Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro	690	695	700
Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe	705	710	715
Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr	725	730	735
Arg Tyr Leu Thr Arg Asn Leu	740		

<210> SEQ ID NO 63
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 63

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	1	5	10	15
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro	20	25	30	
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro	35	40	45	
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro	50	55	60	
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	65	70	75	80
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala	85	90	95	
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly	100	105	110	
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro	115	120	125	
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg	130	135	140	
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly	145	150	155	160
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	165	170	175	
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro	180	185	190	
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly	195	200	205	
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser	210	215	220	
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile				

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225					230					235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
			260					265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
	290					295				300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
			325						330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
		340						345					350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
	355						360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
	370					375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390					395					400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
			405					410						415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
		420						425					430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
	435						440					445			
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
	450					455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
			485					490						495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
		500						505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
	515						520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530				535						540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
		565							570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Thr	Asn	Ser	Val
		580						585					590		
Lys	Leu	Ile	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
	595					600					605				
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
	610					615				620					
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625					630					635					640

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Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 64
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 64

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175
 Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
 180 185 190
 Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
 195 200 205
 Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
 210 215 220
 Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
 225 230 235 240

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Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
			260					265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
	290					295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
305					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
			325						330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340					345					350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355				360						365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
	370					375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
385					390					395					400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
			405					410						415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
			420				425						430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
		435				440					445				
Arg	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
	450					455					460				
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
				485				490						495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
		515				520					525				
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530				535					540					
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545				550					555						560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565					570						575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Thr	Ser	Val	Val
			580					585					590		
Pro	Val	Gly	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile
		595				600					605				
Leu	Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro
610					615					620					
Ile	Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro
625				630					635						640

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Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile
 645 650 655
 Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp
 660 665 670
 Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val
 675 680 685
 Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro
 690 695 700
 Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe
 705 710 715 720
 Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr
 725 730 735
 Arg Tyr Leu Thr Arg Asn Leu
 740

<210> SEQ ID NO 65
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 65

gctcaggcta atactgttcg taatactgct cag

33

<210> SEQ ID NO 66
 <211> LENGTH: 31
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 66

gctcaggata atcatgtcgt actgtgctca g

31

<210> SEQ ID NO 67
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 67

gctcaggaga attctactaa gtctttggct cag

33

<210> SEQ ID NO 68
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 68

gctcaggaga atactgttcg taatatggct cag

33

<210> SEQ ID NO 69
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 69

gctcagggta atgctattcg taatggtgct cag

33

<210> SEQ ID NO 70

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 70

gctcagggta atgctgttaa gccttttgct cag

33

<210> SEQ ID NO 71

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 71

gctcagggta atggtactcg ttgtttgct cag

33

<210> SEQ ID NO 72

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 72

gctcagcata ataatgttcg tcctgttgct cag

33

<210> SEQ ID NO 73

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 73

gctcagttgt tgcgtgataa gtcttctgct cag

33

<210> SEQ ID NO 74

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 74

gctcagatga ataatactgc tgttttggct cag

33

<210> SEQ ID NO 75

<211> LENGTH: 33

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 75

gctcagaata atactactaa gttggctgct cag

33

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<210> SEQ ID NO 76
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 76

gctcagaata atactgttaa gaatttggt cag 33

<210> SEQ ID NO 77
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 77

gctcagtcta atgttattcg taattctgct cag 33

<210> SEQ ID NO 78
<211> LENGTH: 32
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 78

gctcagactt gaagtctcct gctactgctc ag 32

<210> SEQ ID NO 79
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 79

gctcagacta attctgttaa ggctactgct cag 33

<210> SEQ ID NO 80
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 80

gctcagactt ctgttggtcc tgttggtgct cag 33

<210> SEQ ID NO 81
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 81

gctcagtata attctactaa gaatatggct cag 33

<210> SEQ ID NO 82
<211> LENGTH: 33

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 82

gctcaggcta attctgttaa gaattattgct cag          33

<210> SEQ ID NO 83
<211> LENGTH: 33
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

<400> SEQUENCE: 83

gctcagttga atactactaa gcttattgct cag          33

<210> SEQ ID NO 84
<211> LENGTH: 2211
<212> TYPE: DNA
<213> ORGANISM: adeno-associated virus 9

<400> SEQUENCE: 84

atgggtgccg atgggttatct tccagattgg ctccaggaca accttagtga aggaattcgc      60
gagtgggtggg ctttgaagacc tggagcccct caacccaagg caaatcaaca acatcaagac     120
aacgctcgag gtcttgtgct tccgggttac aaataccttg gaccggcaa cggaactcgac     180
aaggggggagc cgggtcaacgc agcagacgcg gcggccctcg agcacgacaa ggcttacgac     240
cagcagctca aggcgggaga caaccggtac ctcaagtaca accacgcoga cggcgagttc     300
caggagcggc tcaaagaaga tacgtctttt gggggcaacc tcgggcgagc agtcttcag      360
gccaaaaaga ggcttcttga acctcttggc ctgggtgagg aagcggctaa gacggctcct     420
ggaaagaaga ggctgttaga gcagctctct caggaaccgc actcctccgc ggggtattggc     480
aaatcgggtg cacagccgcg taaaaagaga ctcaatttcg gtcagactgg cgacacagag     540
tcagtcaccg acctcaacc aatcggagaa cctcccgag cccctcagg tgtgggatct      600
cttacaatgg cttcaggtgg tggcgacca gtggcagaca ataacgaagg tgccgatgga     660
gtgggtagtt cctcgggaaa ttggcattgc gattcccaat ggctggggga cagagtcac      720
accaccagca cccgaacctg ggccctgccc acctacaaca atcacctcta caagcaaac      780
tccaacagca catctggagg atcttcaaat gacaacgcct acttcggcta cagcaccccc     840
tgggggtatt ttgaattcaa cagattccac tgccaattct caccacgtga ctggcagcga     900
ctcatcaaca acaactgggg attccggcct aagcgactca acttcaagct cttcaacatt     960
caggtcaaaag aggttacgga caacaatgga gtcaagacca tcgccaataa ccttaccagc    1020
acggtcacag tcttcacgga ctcagactat cagctcccg acgtgctcgg gtcgggtcac    1080
gagggtgcc tcccgcggtt cccagcggac gttttcatga ttcctcagta cgggtatctg    1140
acgcttaatg atggaagcca ggcggtgggt cgttcgtcct tttactgcct ggaatatctc    1200
ccgtcgcaaa tgctaagaac gggtaacaac ttccagttca gctacgagtt tgagaacgta    1260
cttttcata gcagctacgc tcacagccaa agcctggacc gactaatgaa tccactcacc    1320
gaccaatact tgtactatct ctcaaagact attaacggtt ctggacagaa tcaacaaacg    1380
ctaaaattca gtgtggccgg acccagcaac atggctgtcc agggaagaaa ctacatacct    1440

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ggacccagct accgacaaca acgtgtctca accactgtga ctcaaaacaa caacagcgaa 1500
tttgcttggc ctggagcttc ttcttgggct ctcaatggac gtaatagctt gatgaatcct 1560
ggacctgcta tggccagcca caaagaagga gaggaccgtt tctttccttt gtctggatct 1620
ttaatttttg gcaaacagg aactggaaga gacaacgtgg atgcggacaa agtcatgata 1680
accaacgaag aagaaattaa aactactaac ccggtagcaa cggagtccta tggacaagtg 1740
gccacaaacc accagagtgc ccaagcacag gcgcagaccg gctgggttca aaaccaagga 1800
atacttccgg gtatggtttg gcaggacaga gatgtgtacc tgcaaggacc catttgggcc 1860
aaaattcctc acacggacgg caactttcac ccttctccgc tgatgggagg gtttggaatg 1920
aagcaccgcg ctctcagat cctcatcaaa aacacacctg tacctgcgga tcctccaacg 1980
gccttcaaca aggacaagct gaactcttcc atcaccagc attctactgg ccaagtcagc 2040
gtggagatcg agtgggagct gcagaaggaa aacagcaagc gctggaaccc ggagatccag 2100
tacacttcca actattacaa gtctaataat gttgaatttg ctgttaatac tgaagggtga 2160
tatagtgaac ccgcgcccat tggcaccaga tacctgactc gtaatctgta a 2211

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<210> SEQ ID NO 85
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 85

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Lys Asp Asn Thr Pro Gly Arg
1             5

```

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<210> SEQ ID NO 86
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthesized

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<400> SEQUENCE: 86

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Asp Gly Ala Ala Thr Lys Asn
1             5

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What is claimed is:

1. An AAV capsid protein comprising a sequence as provided in Table 1 or FIG. 1.

2. The AAV capsid protein of claim 1, wherein the sequence is selected from

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		AQGNNAIRNGAQ,	(SEQ ID NO: 6)
		AQGNNAVKPFAQ,	(SEQ ID NO: 7)
		AQGNNGTRPTAQ,	(SEQ ID NO: 8)
AQANTVRNTAQ,	(SEQ ID NO: 2)	AQHNNVRPVAQ,	(SEQ ID NO: 9)
AQDNHVRTVAQ,	(SEQ ID NO: 3)	AQLLRDKSSAQ,	(SEQ ID NO: 10)
AQENSTKSMAQ,	(SEQ ID NO: 4)	AQMNTAVLAQ,	(SEQ ID NO: 11)
AQENTVRNMAQ,	(SEQ ID NO: 5)		

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AQNNTTKLAAQ,	(SEQ ID NO: 12)
AQNNTVKNLAAQ,	(SEQ ID NO: 13)
AQSNVIRNSAQ,	(SEQ ID NO: 14)
AQTLKSPATAQ,	(SEQ ID NO: 15)
AQTNSVKLIAQ,	(SEQ ID NO: 16)
AQTSVVPVGAQ,	(SEQ ID NO: 15)
AQYNSTKNMAQ,	(SEQ ID NO: 18)
AQANSVKINIAQ, and	(SEQ ID NO: 19)
AQLNNTKPIDG.	(SEQ ID NO: 20)

3. An AAV capsid protein comprising an insertion sequence selected from ANTVRNT (SEQ ID NO: 21), DNHVRTV (SEQ ID NO: 22), ENSTKSM (SEQ ID NO: 23), ENTVRNM (SEQ ID NO: 24), GNAIRNG (SEQ ID NO: 25), GNAVKPF (SEQ ID NO: 26), GNGTRPT (SEQ ID NO: 27), HNNVRPV (SEQ ID NO: 28), LLRDKSS (SEQ ID NO: 29), MNNTAVL (SEQ ID NO: 30), NNTTKLA (SEQ ID NO: 31), NNTVKNL (SEQ ID NO: 32), SNVIRNS (SEQ ID NO: 33), TLKSPAT (SEQ ID NO: 34), TNSVKLI (SEQ ID NO: 35), TSVVPVG (SEQ ID NO: 36), YNSTKNM (SEQ ID NO: 37), and ANSVKNI (SEQ ID NO: 38).

4. The AAV capsid protein of any of claims 1-3 wherein the AAV is AAV9.

5. The AAV capsid protein of any of claims 1-4 wherein the AAV is provided in SEQ ID NO: 1

6. The AAV capsid protein of any of claims 1-5 wherein the insertion is between amino acid 588 and amino acid 589.

7. The AAV capsid protein of any one of claims 1-6, wherein 60 copies of the AAV capsid protein are assembled into the AAV capsid.

8. The AAV capsid protein of any of claims 1-7, wherein the AAV capsid protein is present in VP1, VP2, and VP3 of the AAV capsid.

9. The AAV capsid protein of any of claims 1-8, further characterized by an increased transduction enrichment when measured in a CNS in a subject when delivered to the subject systemically.

10. The AAV capsid protein of any of claims 3-9, wherein the AAV capsid protein further comprises an amino acid substitution comprising A587D and/or Q588G.

11. An AAV capsid comprising an AAV capsid protein of any of claims 1-10.

12. The AAV capsid of claim 11 wherein the AAV capsid is chimeric.

13. The AAV capsid of any of claims 11-12 that is isolated and purified.

14. The AAV capsid of any of claims 11-13 formulated as a pharmaceutical formulation for systemic administration to

treat a disease or a condition of the CNS, the pharmaceutical formulation further comprising a pharmaceutically acceptable carrier.

15. The AAV capsid protein of any one of claims 1-10, wherein the capsid protein is expressed in the CNS.

16. The AAV capsid protein of claim 15, wherein the CNS comprises a cell-type selected from a neuron, an oligodendrocyte, an astrocyte, and a brain vascular cell.

17. The AAV capsid protein of any one of claims 15-16, wherein the CNS comprises a tissue that is selected from brain, thalamus, cortex, putamen, lateral ventricles, medulla, the pons, the amygdala, the motor cortex, caudate, hypothalamus, striatum, ventral midbrain, neocortex, basal ganglia, hippocampus, thalamus, cerebrum, cerebellum, brain stem, and spinal cord.

18. A nucleic acid sequence encoding the peptide of any of claims 1-10, selected from the group consisting of SEQ ID NOs 67-85.

19. A nucleic acid sequence encoding a peptide sequence as provided in Table 1 or FIG. 1.

20. A recombinant vector comprising a nucleic acid encoding the AAV capsid protein of any one of claims 1-10.

21. A kit comprising:

- a) a first vector comprising the recombinant vector of claim 20;
- b) a second vector encoding a helper virus protein; and
- c) a third vector comprising a therapeutic nucleic acid encoding a therapeutic gene expression product.

22. A method of treating a disease or condition in a subject comprising administering a therapeutically effective amount of a pharmaceutical formulation comprising the AAV capsid protein of any one of claims 1-10.

23. The method of claim 22, wherein the disease or the condition is a disease or a condition of a CNS of the subject.

24. A method of manufacturing a recombinant AAV particle from the AAV capsid of any one of claims 11-13, the method comprising:

- a. introducing into a cell a nucleic acid comprising:
 - i.a first nucleic acid sequence encoding a therapeutic gene expression product;
 - ii.a second nucleic acid sequence encoding a recombinant viral genome comprising a capsid (Cap) gene modified to express the AAV capsid of any one of claims 11-12; and
 - iii.a third nucleic acid sequence encoding an AAV helper virus genome; and
- b. assembling the recombinant AAV particle, the recombinant AAV particle comprising the AAV capsid encapsidating the first nucleic acid.

25. The AAV capsid protein of any of claims 1-10 and 15-17 wherein the AAV capsid protein is characterized by an increased transduction enrichment when measured in a tissue in a subject when delivered to the subject systemically.

26. The AAV capsid protein of any of claims 1-10 and 15-17 wherein the tissue is brain tissue.

27. An AAV particle comprising an AAV capsid protein of any of claims 1-10 and 15-17 and a viral genome.

28. A peptide comprising an amino acid sequence selected from ANTVRNT (SEQ ID NO: 21), DNHVRTV (SEQ ID NO: 22), ENSTKSM (SEQ ID NO: 23), ENTVRNM (SEQ ID NO: 24), GNAIRNG (SEQ ID NO: 25), GNAVKPF (SEQ ID NO: 26), GNGTRPT (SEQ ID NO: 27), HNNVRPV (SEQ ID NO: 28), LLRDKSS (SEQ ID NO:

29), MNNTAVL (SEQ ID NO: 30), NNTTKLA (SEQ ID NO: 31), NNTVKNL (SEQ ID NO: 32), SNVIRNS (SEQ ID NO: 33), TLKSPAT (SEQ ID NO: 34), TNSVKLI (SEQ ID NO: 35), TSVVPVG (SEQ ID NO: 36), YNSTKNM (SEQ ID NO: 37), and ANSVKNI (SEQ ID NO: 38).

29. A recombinant adeno-associated virus (rAAV) comprising a capsid comprising any one of

ANTVRNT,	(SEQ ID NO: 21)
DNHVRTV,	(SEQ ID NO: 22)
ENSTKSM,	(SEQ ID NO: 23)
ENTVRNM,	(SEQ ID NO: 24)
GNAIRNG,	(SEQ ID NO: 25)
GNAVKPF,	(SEQ ID NO: 26)
GNGTRPT,	(SEQ ID NO: 27)
HNNVRPV,	(SEQ ID NO: 28)
LLRDKSS,	(SEQ ID NO: 29)
MNNTAVL,	(SEQ ID NO: 30)
NNTTKLA,	(SEQ ID NO: 31)
NNTVKNL,	(SEQ ID NO: 32)
SNVIRNS,	(SEQ ID NO: 33)
TLKSPA,	(SEQ ID NO: 34)
TNSVKLI,	(SEQ ID NO: 35)
TSVVPVG,	(SEQ ID NO: 36)
YNSTKNM,	(SEQ ID NO: 37)
and	
ANSVKNI.	(SEQ ID NO: 38)

30. A recombinant adeno-associated virus (rAAV) comprising a capsid comprising any one of

AQANTVRNTAQ,	(SEQ ID NO: 2)
AQDNHVRTVAQ,	(SEQ ID NO: 3)
AQENSTKSMAQ,	(SEQ ID NO: 4)
AQENTVRNMAQ,	(SEQ ID NO: 5)
AQGNAIRNGAQ,	(SEQ ID NO: 6)
AQGNAVKPFAQ,	(SEQ ID NO: 7)
AQGNGTRPTAQ,	(SEQ ID NO: 8)
AQHNNVRPVAQ,	(SEQ ID NO: 9)
AQLLRDKSSAQ,	(SEQ ID NO: 10)
AQMNTAVLAQ,	(SEQ ID NO: 11)
AQNNTTKLAAQ,	(SEQ ID NO: 12)
AQNNTVKNLAQ,	(SEQ ID NO: 13)
AQSNVIRNSAQ,	(SEQ ID NO: 14)
AQTLKSPATAQ,	(SEQ ID NO: 15)
AQTNSVKLIAQ,	(SEQ ID NO: 16)
AQTSVVPVGAQ,	(SEQ ID NO: 15)
AQYNSTKNMAQ,	(SEQ ID NO: 18)
AQANSVKNIAQ,	(SEQ ID NO: 19)
and	
AQLNTTKPIDG.	(SEQ ID NO: 20)

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